

Graph models and hypothesis testing

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what I'd get anyway?

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- An analyst finds their organisation's email network has "unexpectedly high clustering" — colleagues of colleagues email each other a lot. Management concludes there are strong informal teams. Is that conclusion justified?

Is what I'm seeing in my network surprising, or is it
what I'd get anyway?

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- In a trade network, one country sits in the middle of everything, and commentators conclude it is structurally special — a deliberate broker. Or does it just trade a lot?

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- In a trade network, one country sits in the middle of everything, and commentators conclude it is structurally special — a deliberate broker. Or does it just trade a lot?
- In both cases the observation is real — but the explanation is only warranted if the pattern exceeds our “expectation”.

The null-model ladder

Rung	Null model	What it holds fixed	Failing it tells you...
1	Erdős–Rényi	number of nodes & edges	"not maximally random" (almost everything fails this)
2	Configuration model	every node's degree	there is structure beyond degree heterogeneity
3	DC-SBM	degrees and group structure	structure beyond blocks + degrees → Wednesday's session
—	BESTest	a node attribute you have	this attribute is (not) associated with the structure

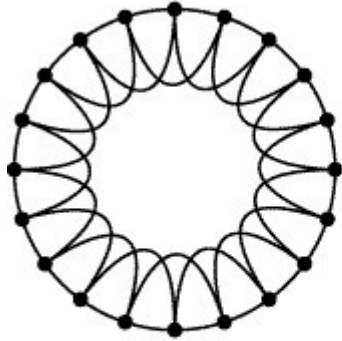
Which rung do you need?

- Do you have a specific count/degree you're questioning?
→ Rung 1 (ER / binomial test)
- Is a global property (clustering, assortativity, degree distribution shape) surprising relative to a random graph?
→ Rung 2 (configuration-model null test)
- Do you suspect group/block structure, even without knowing the groups?
→ Rung 3 (DC-SBM)
- Do you have a labelled attribute you want to test for relevance?
→ BESTest

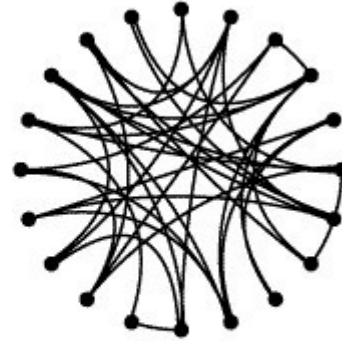
Don't skip to the most complex model — the point is knowing which baseline your result actually needs to beat.

Regular and random graphs

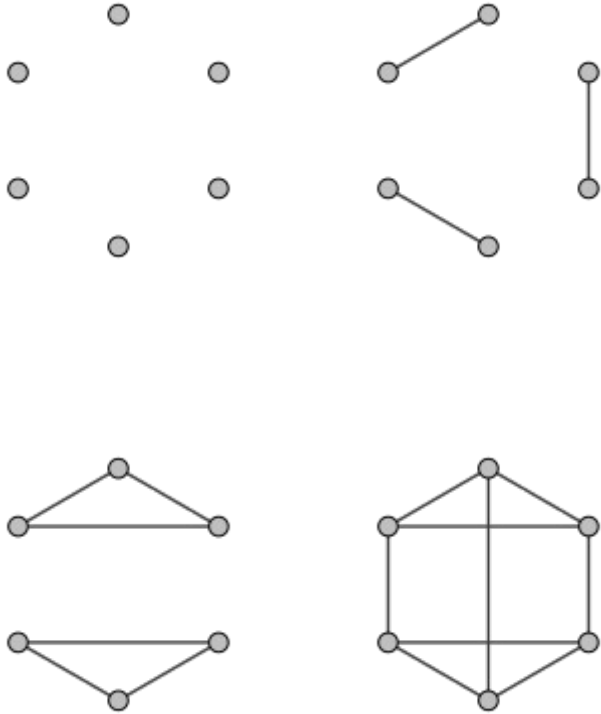
Regular



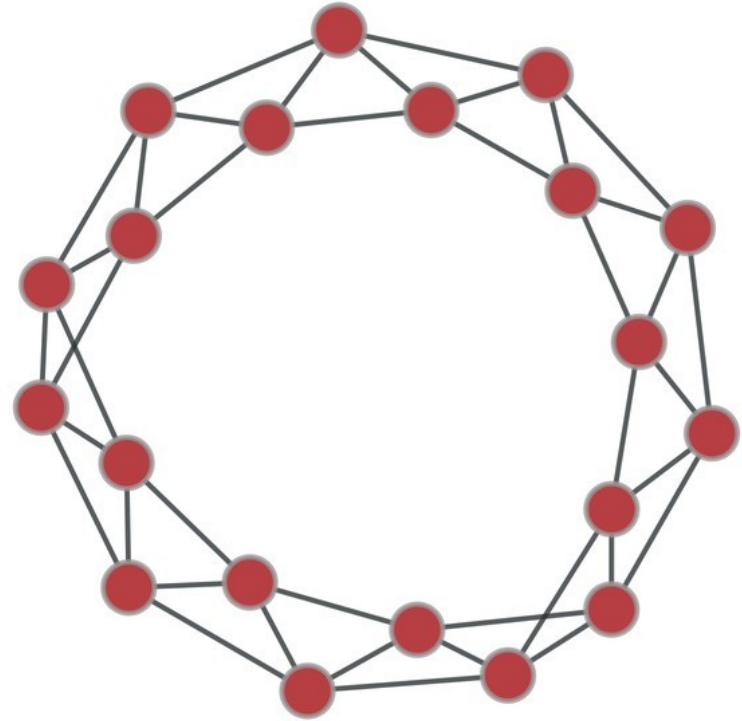
Random



Regular graphs

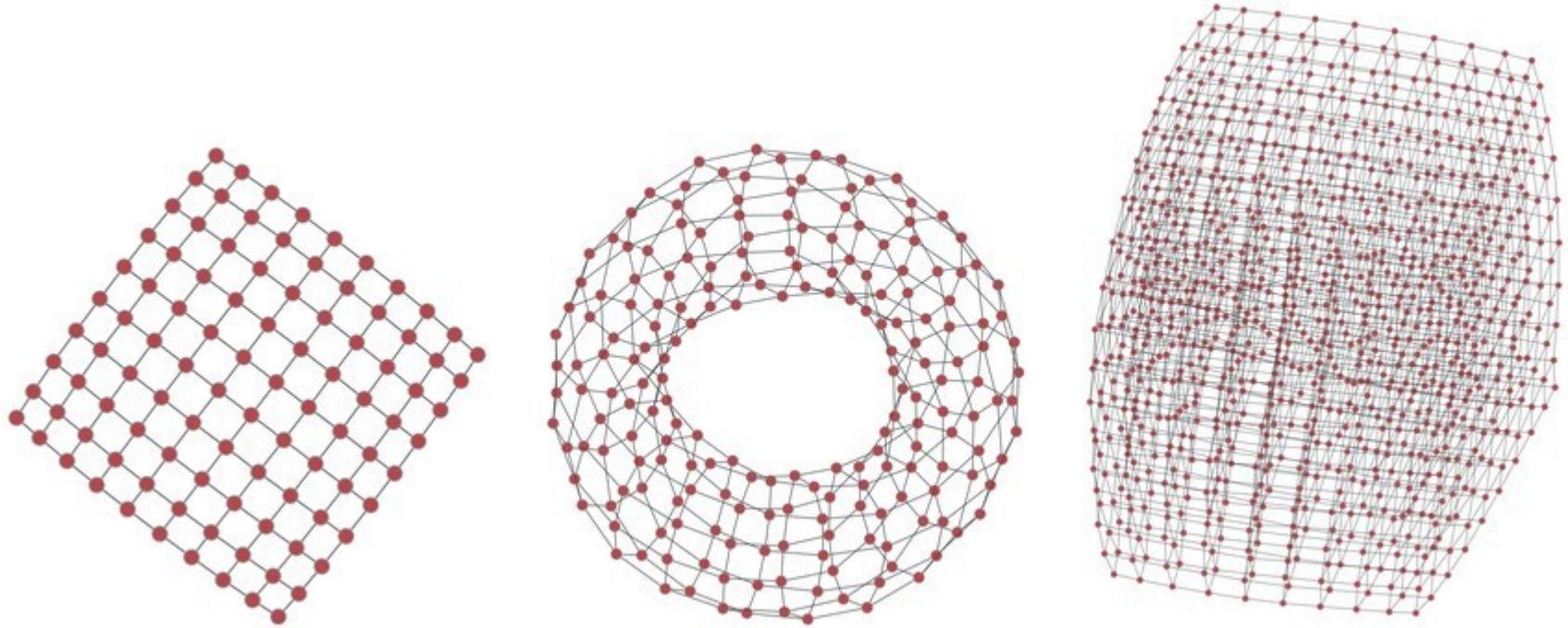


Regular graphs with 0 – 3 degree nodes



Regular Ring Lattice

Lattice graphs



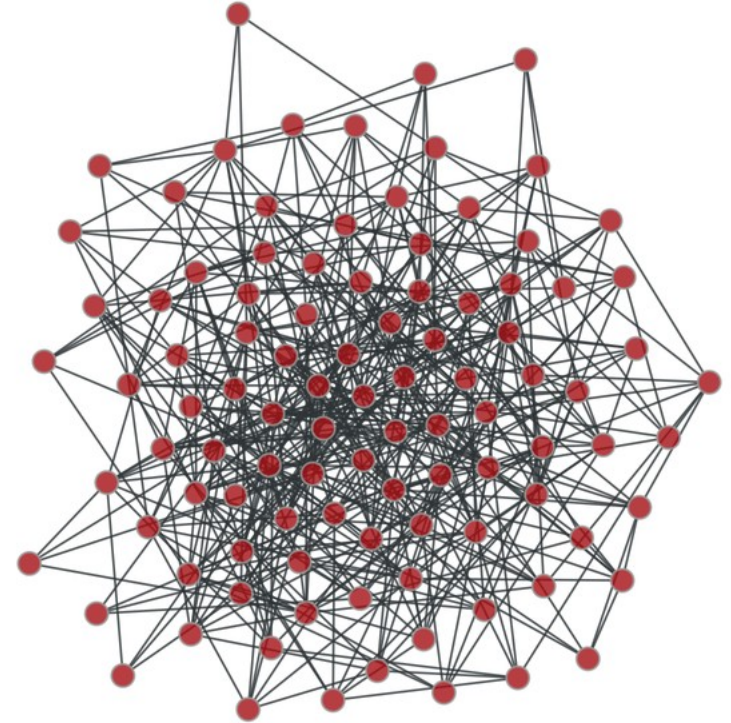
Every time we generate regular/lattice graphs we get the same output

Random graphs

The Erdos-Renyi model

Specified by a number of nodes, n ,
and either:

- a number of edges, m
- a probability of connection, p



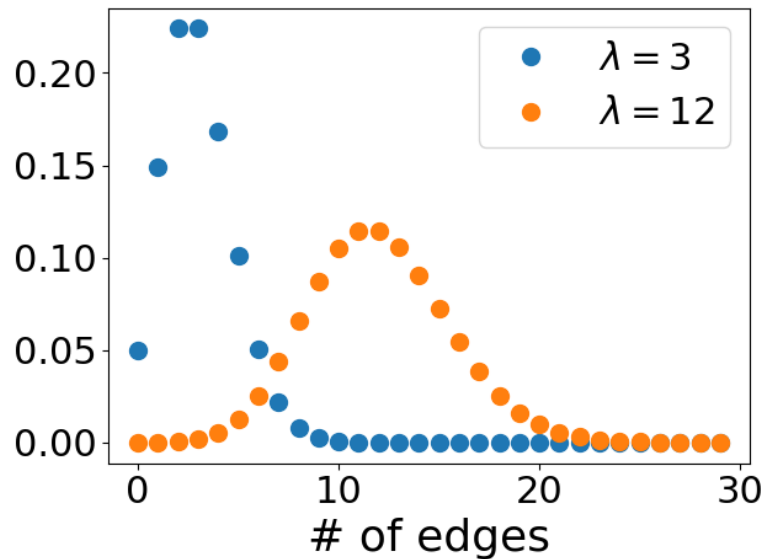
All edges are equally likely to exist

Properties of random graphs

- Locally tree-like
- Large connected component for mean degree greater than 1
- Short path lengths

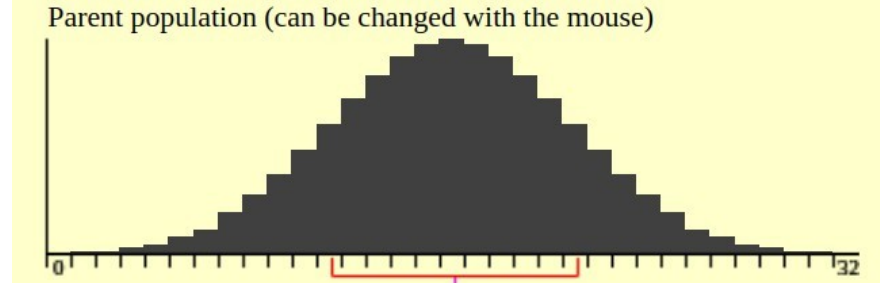
When is the ER not so good?

- Degree distribution
- Clustering



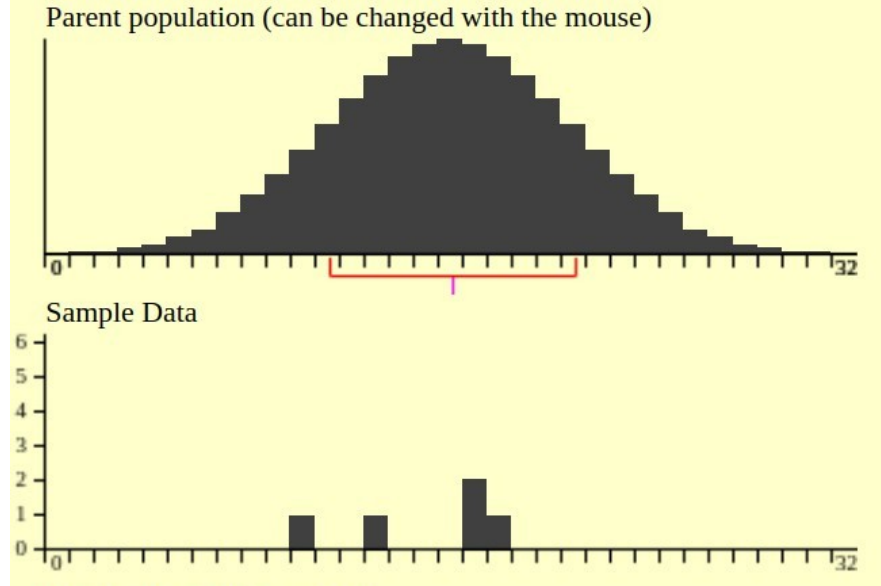
Null hypothesis testing

Population distribution



Mean = 16.00
Sd = 5.00

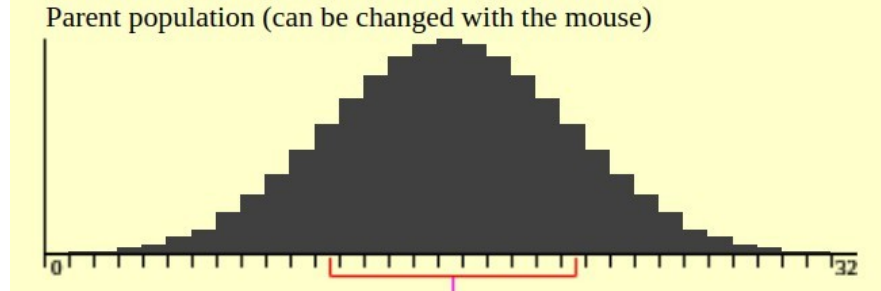
Population distribution



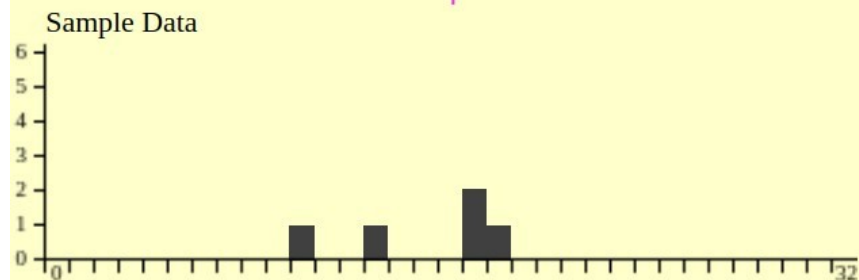
Mean = 16.00
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One sample
(size = 5)

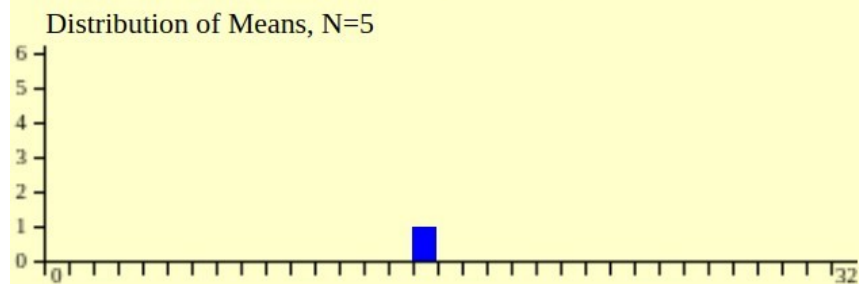
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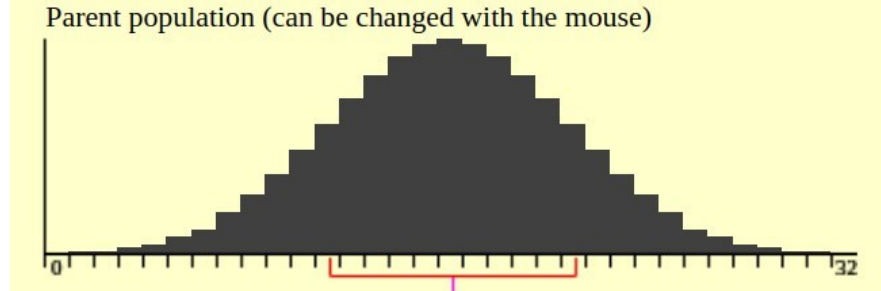
One sample
(size = 5)



Distribution of means
(1 sample)

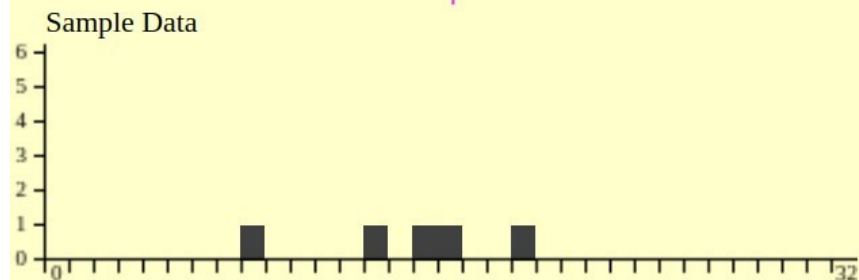


Population distribution

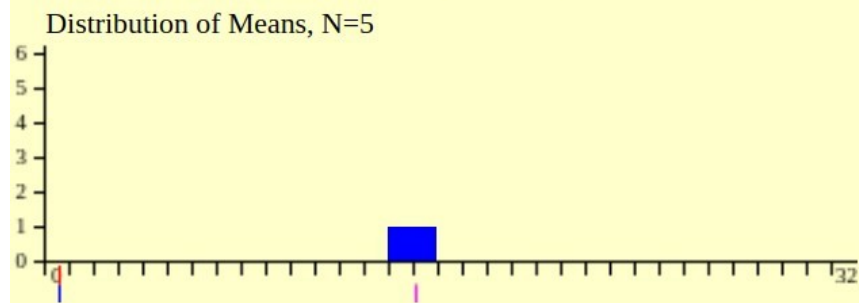


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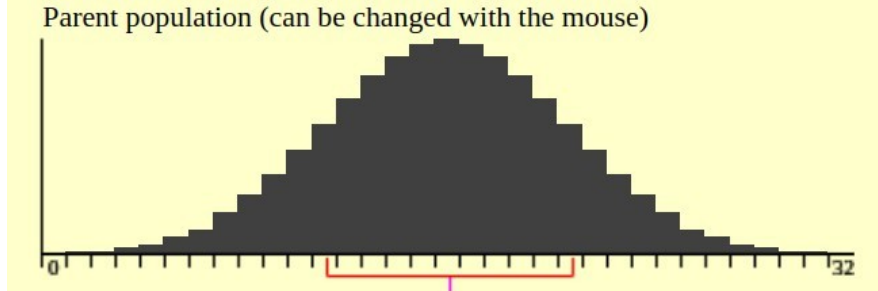
One sample
(size = 5)



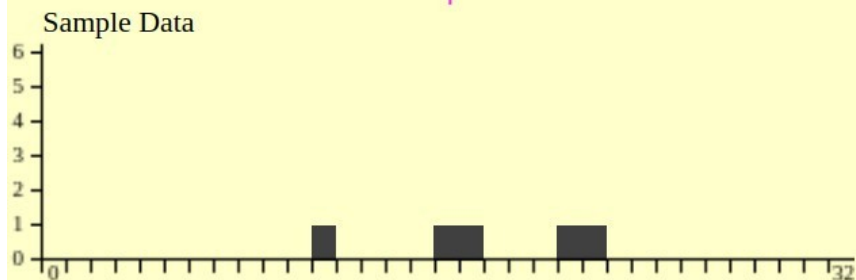
Distribution of means
(2 samples)



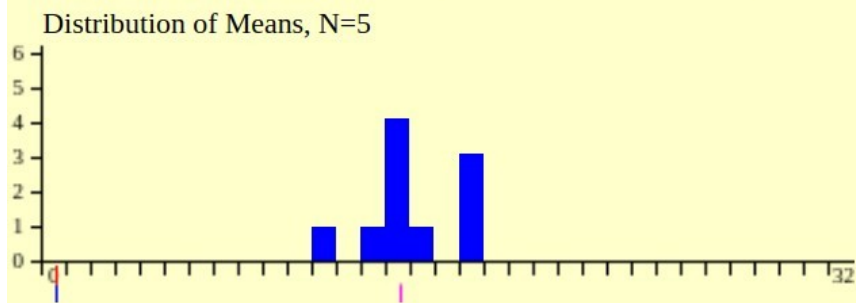
Population distribution



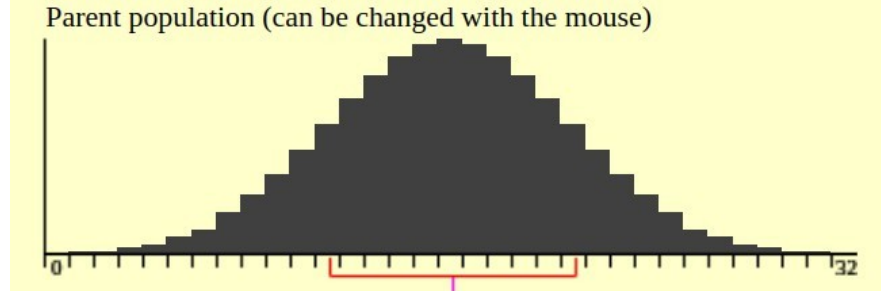
One sample
(size = 5)



Distribution of means
(10 samples)

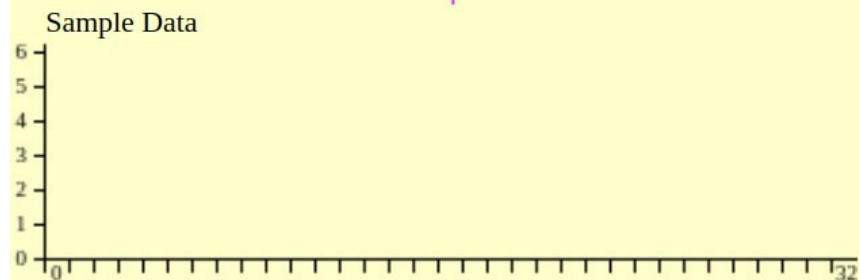


Population distribution

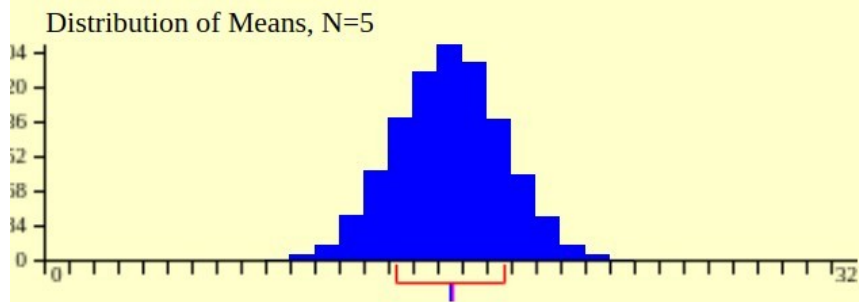


Mean = 16.00
Sd = 5.00

One sample
(size = 5)

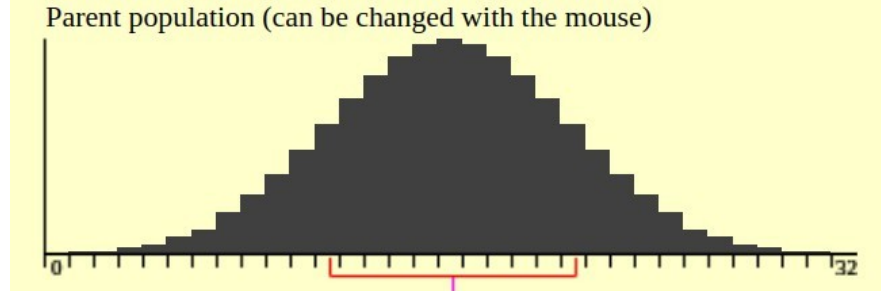


Distribution of means
(10,000 samples)



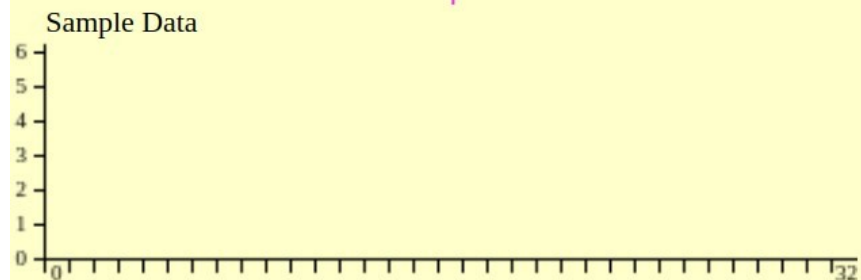
Mean = 15.99
Sd = 2.25

Population distribution

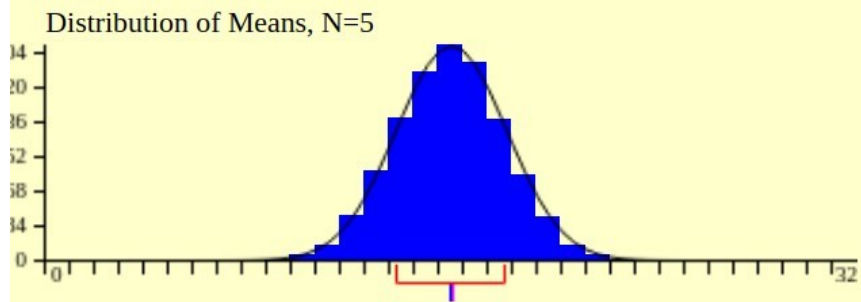


Mean = 16.00
Sd = 5.00

One sample
(size = 5)



Distribution of means
(10,000 samples)



Mean = 15.99
Sd = 2.25

Central limit theorem

The infamous P-value

P-VALUE

The probability, computed assuming that H_0 is true, that the test statistic would take a value as extreme or more extreme than that actually observed is called the ***P*-value** of the test. The smaller the *P*-value, the stronger the evidence against H_0 provided by the data.

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Definition, pg 405
Introduction to the Practice of Statistics, Fifth Edition
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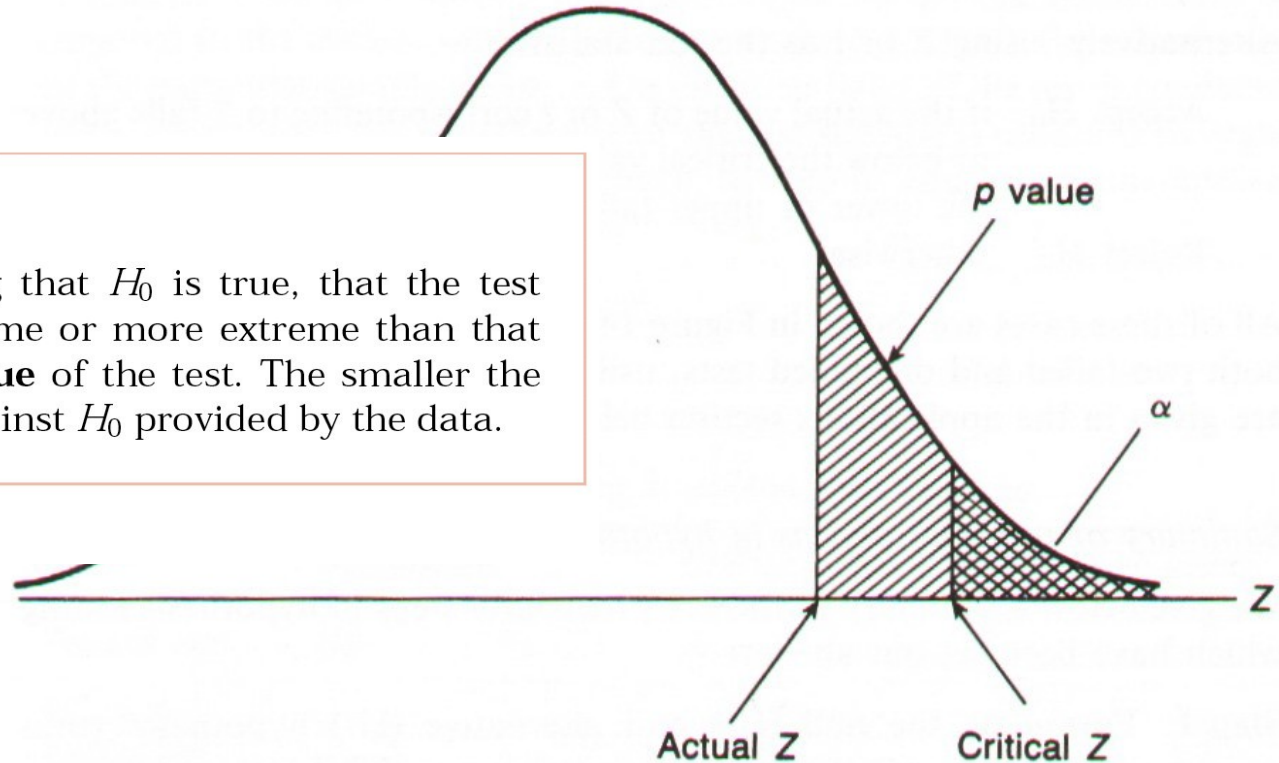


Figure 14.2 Comparison of p values and critical values of Z in a one-tailed test

The infamous P-value

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If $p\text{-value} < \alpha$,
then we reject the
null hypothesis

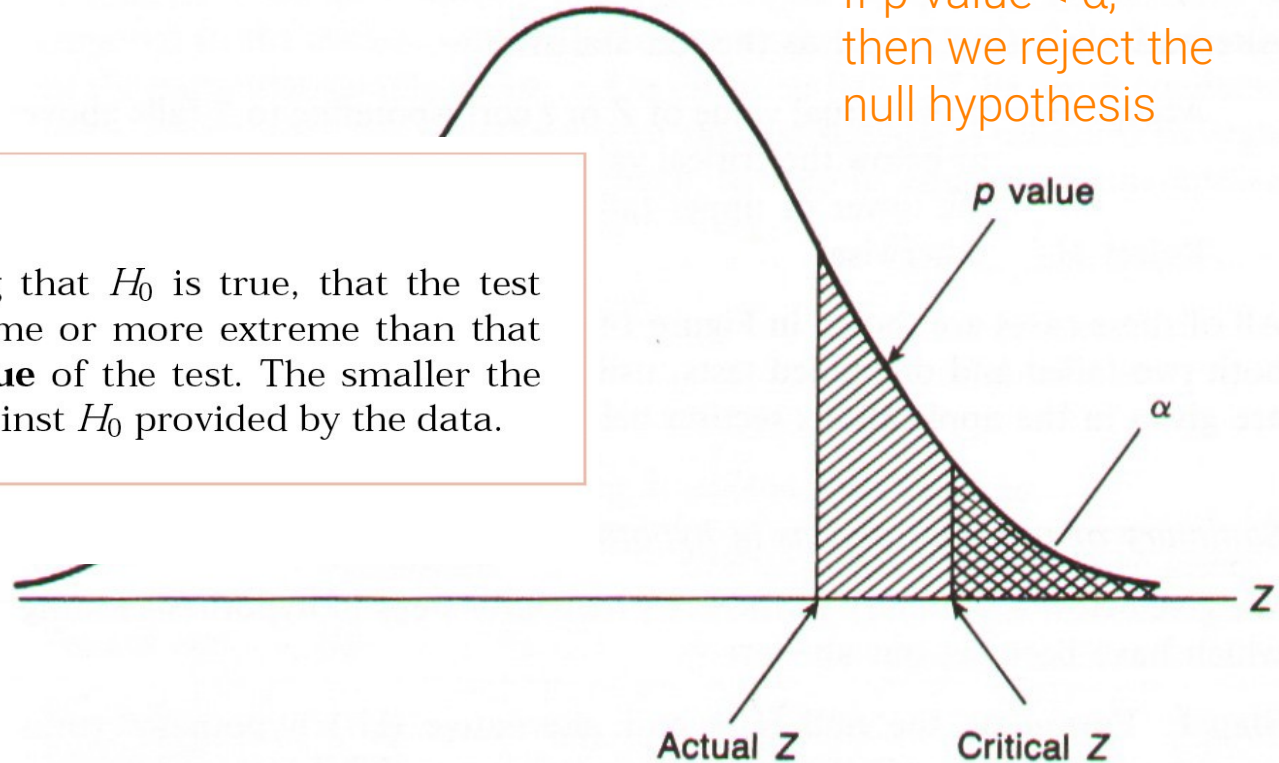


Figure 14.2 Comparison of p values and critical values of Z in a one-tailed test

For networks we can use graph models as a null model

- Does the network appear to be significantly different from a random graph?
- Can specific properties of the observed network (e.g., clustering coefficient) be explained by a particular generative process?
- Two approaches to create samples:
 - permute edges/nodes in a way that is consistent with the graph model
 - “fit” a model to an observed network and generate networks from it

The null-model ladder – you are here

Rung	Null model	What it holds fixed	Failing it tells you...
1	Erdős–Rényi	number of nodes & edges	“not maximally random” (almost everything fails this)
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Practical Part 1

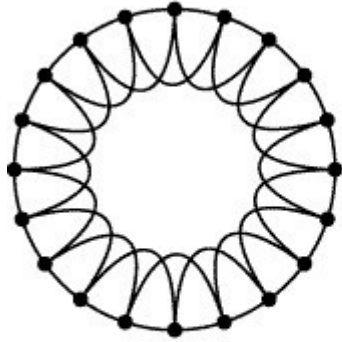
Small-world networks

It's a network after all

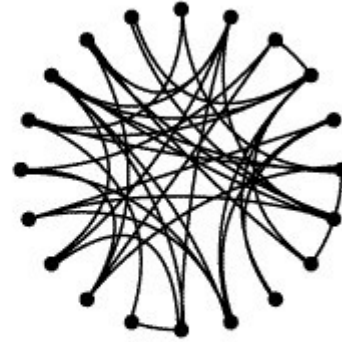
Small-world networks

It's a network after all

Regular



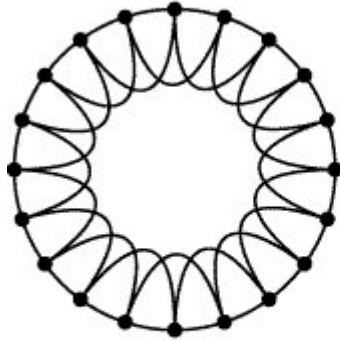
Random



Small-world networks

It's a network after all

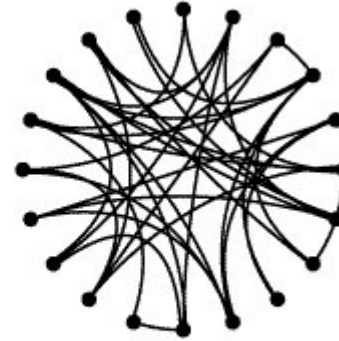
Regular



(triangles)

High **clustering coefficient**

Random

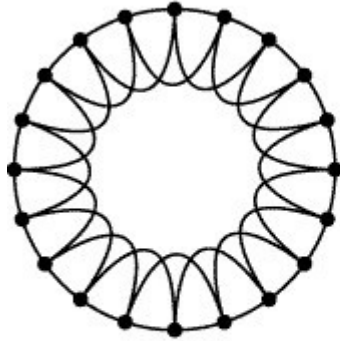


Low **clustering coefficient**

Small-world networks

It's a network after all

Regular



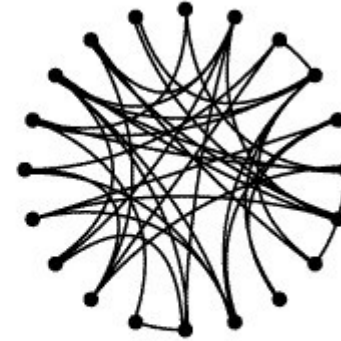
(triangles)

High **clustering coefficient**

(shortest
paths)

High mean **geodesic path**

Random



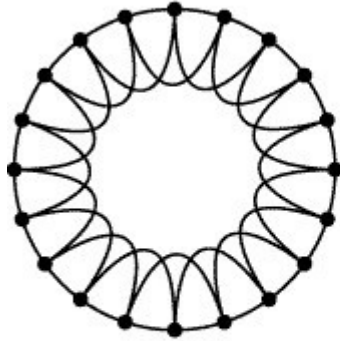
Low **clustering coefficient**

Low mean **geodesic path**

Small-world networks

It's a network after all

Regular



(triangles)

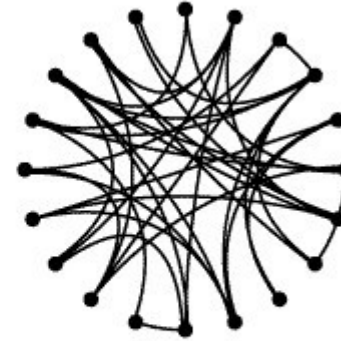
High clustering coefficient

(shortest
paths)

High mean geodesic path

Real-world
networks

Random

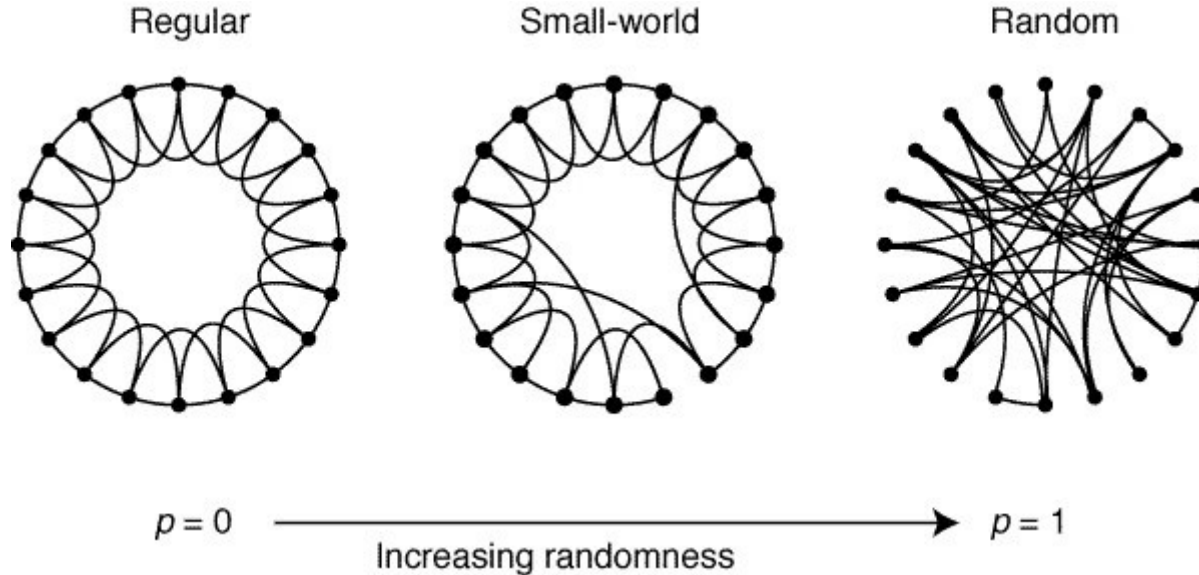


Low clustering coefficient

Low mean geodesic path

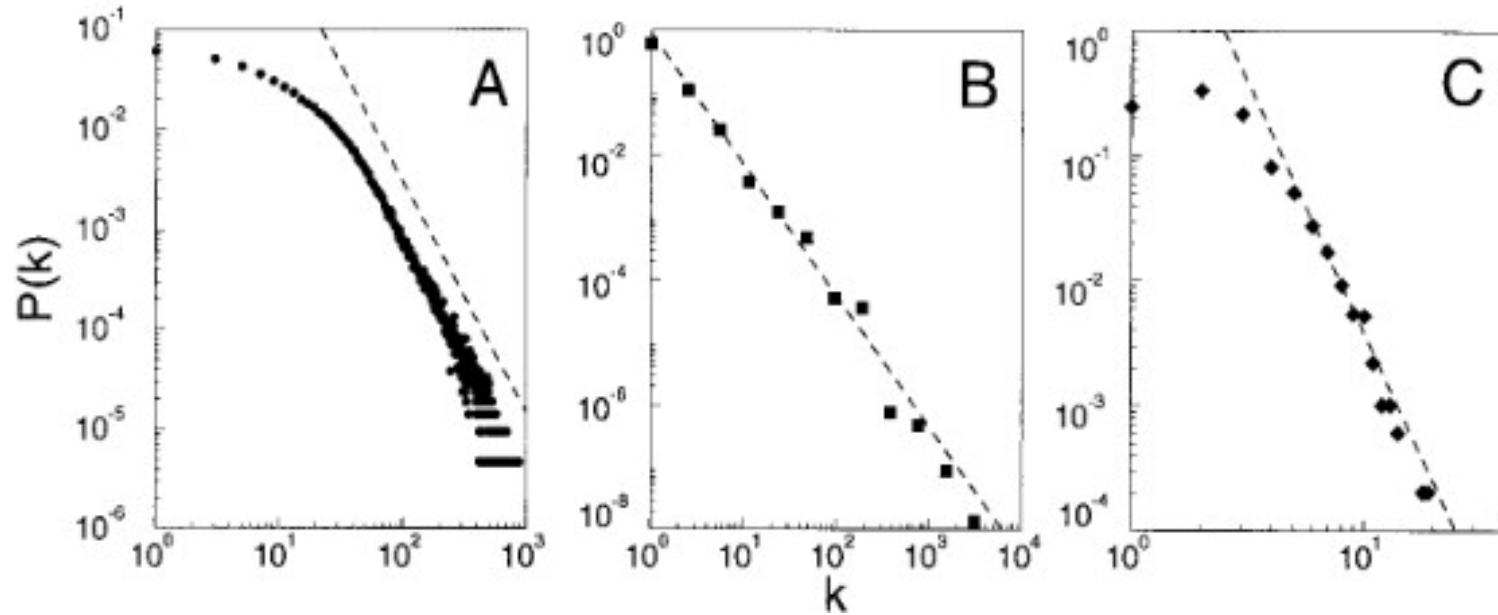
Small-world networks

It's a network after all

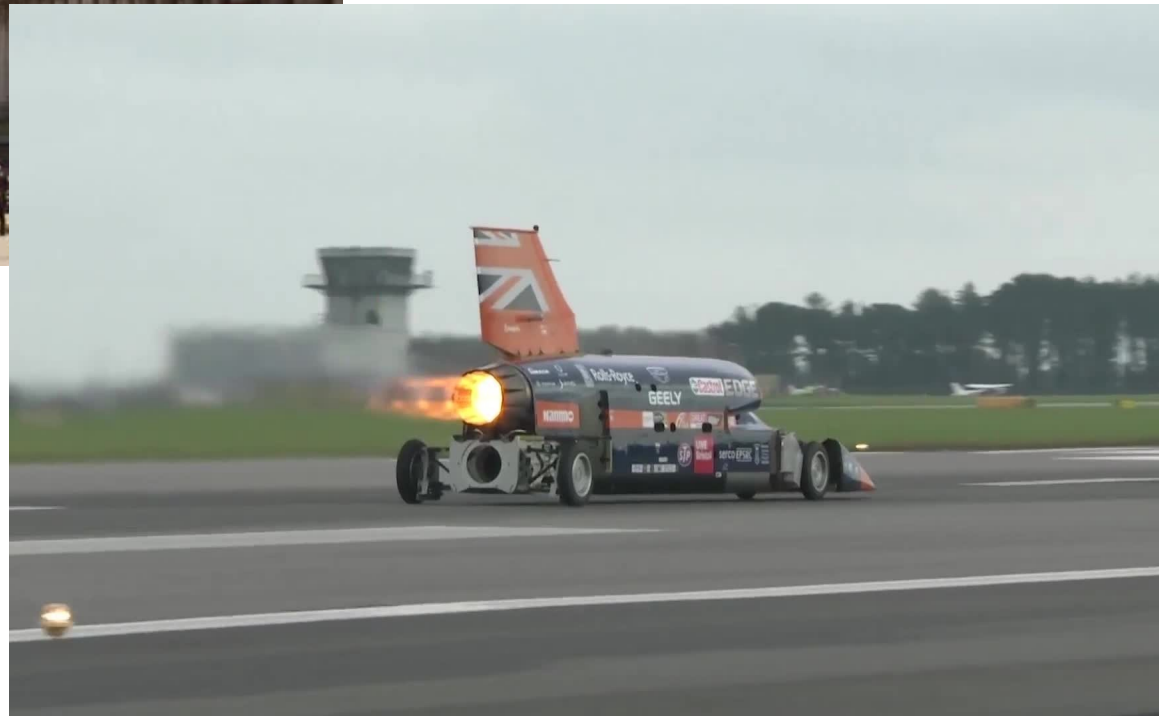


"Scale-free" networks

"Scale-free" networks







The Price model

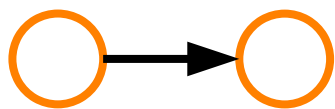
For undirected networks, this model is known as the Barabasi-Albert model

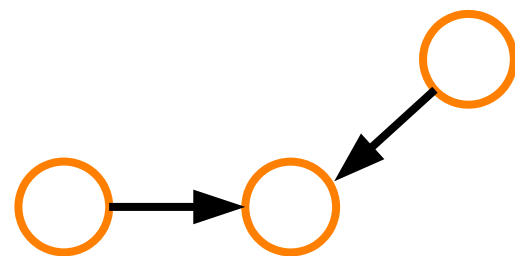
Add nodes to a network one at a time.

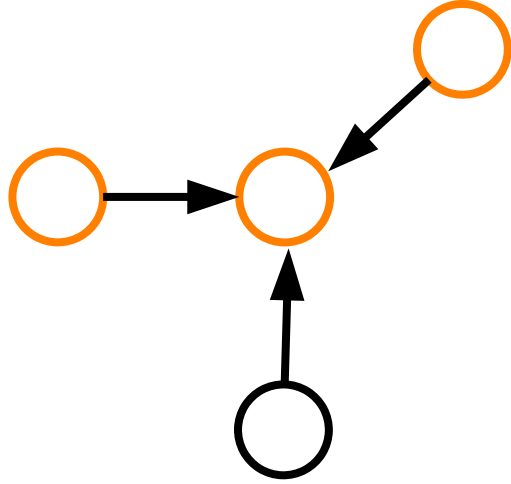
Connect to existing nodes with probability **proportional to their degree**

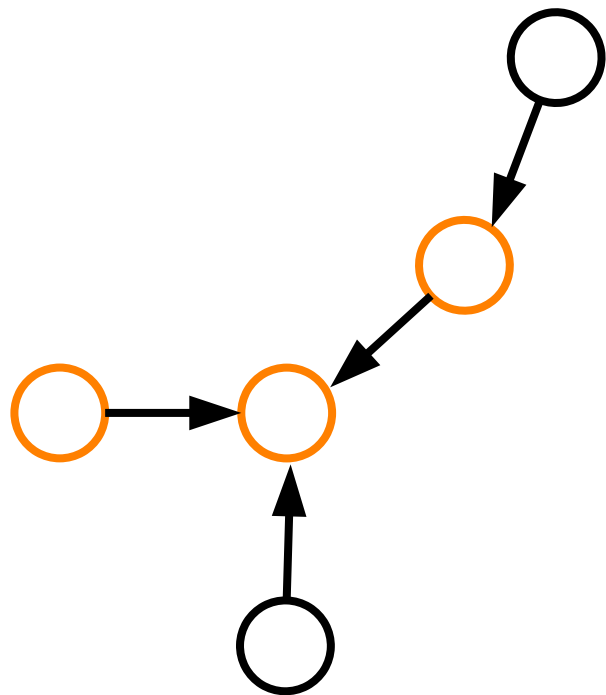
$$p_i = \frac{k_i}{\sum_j k_j},$$

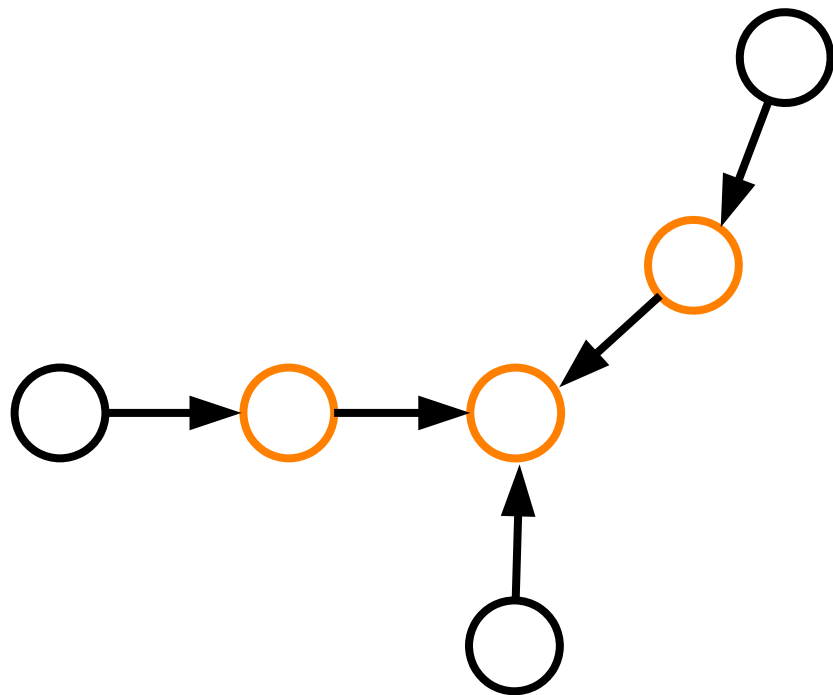


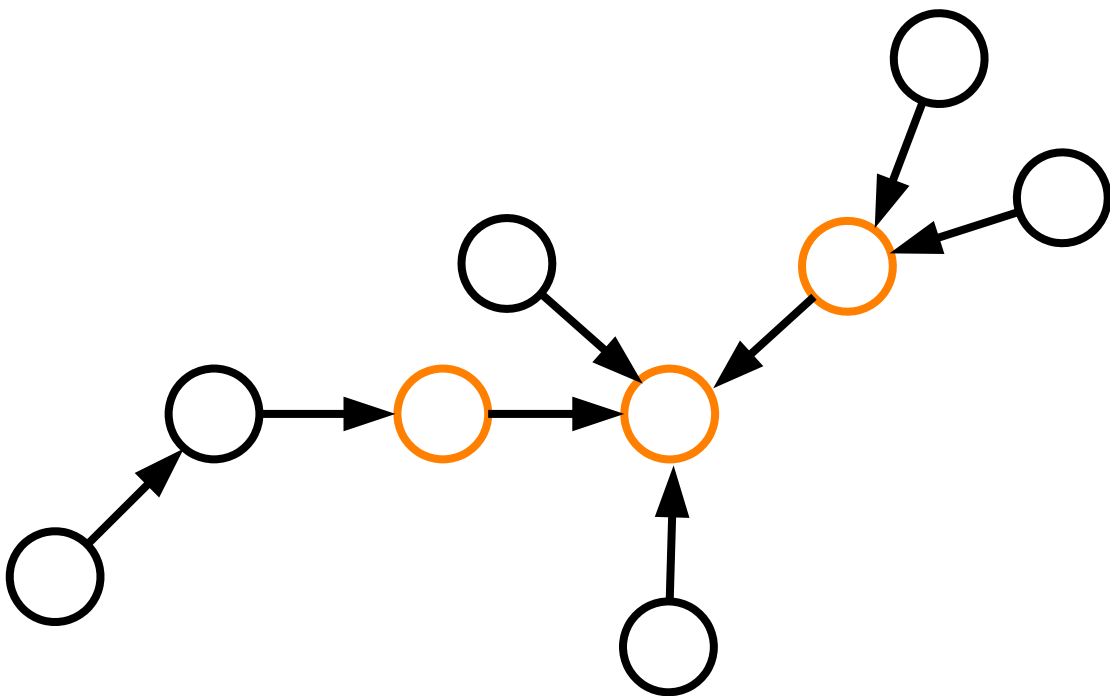


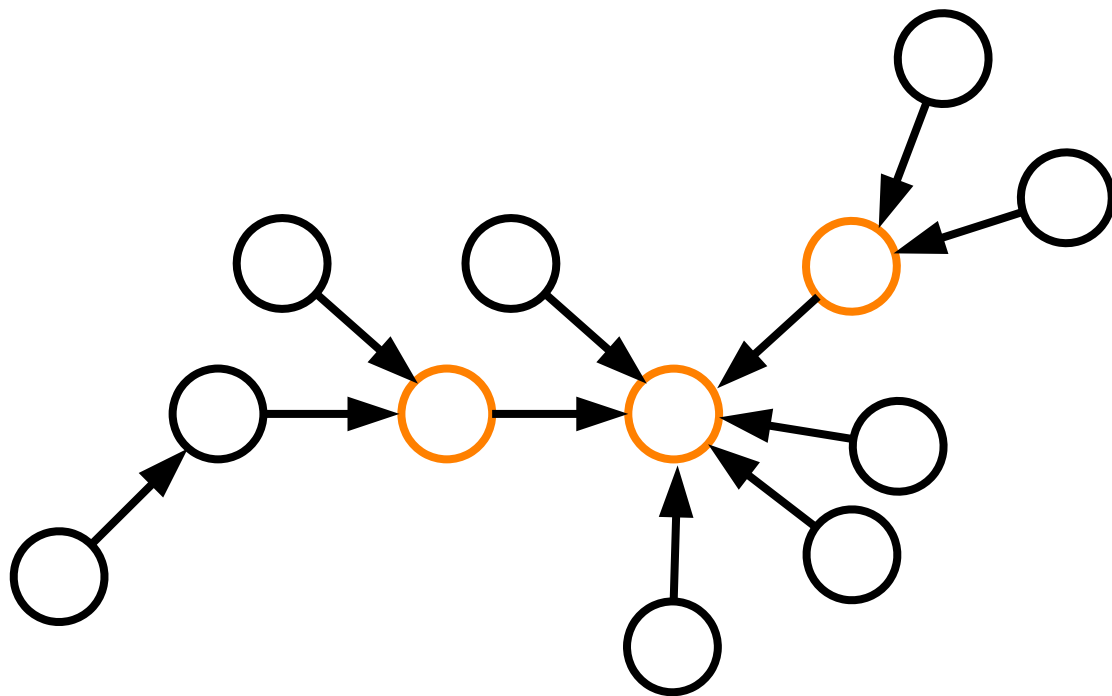


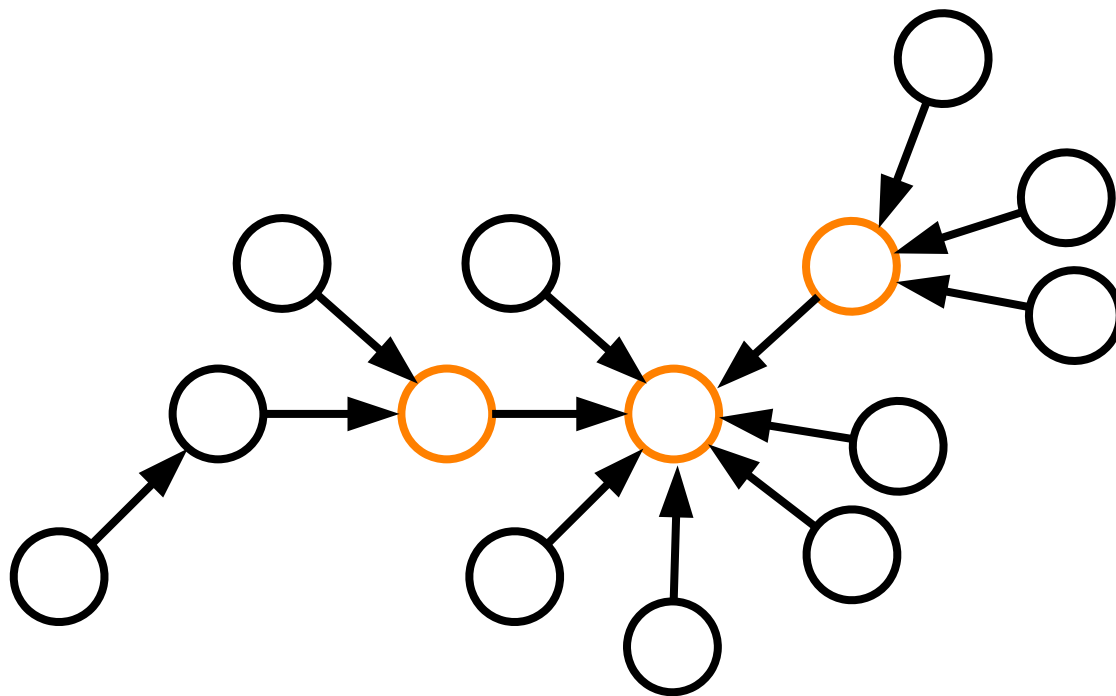




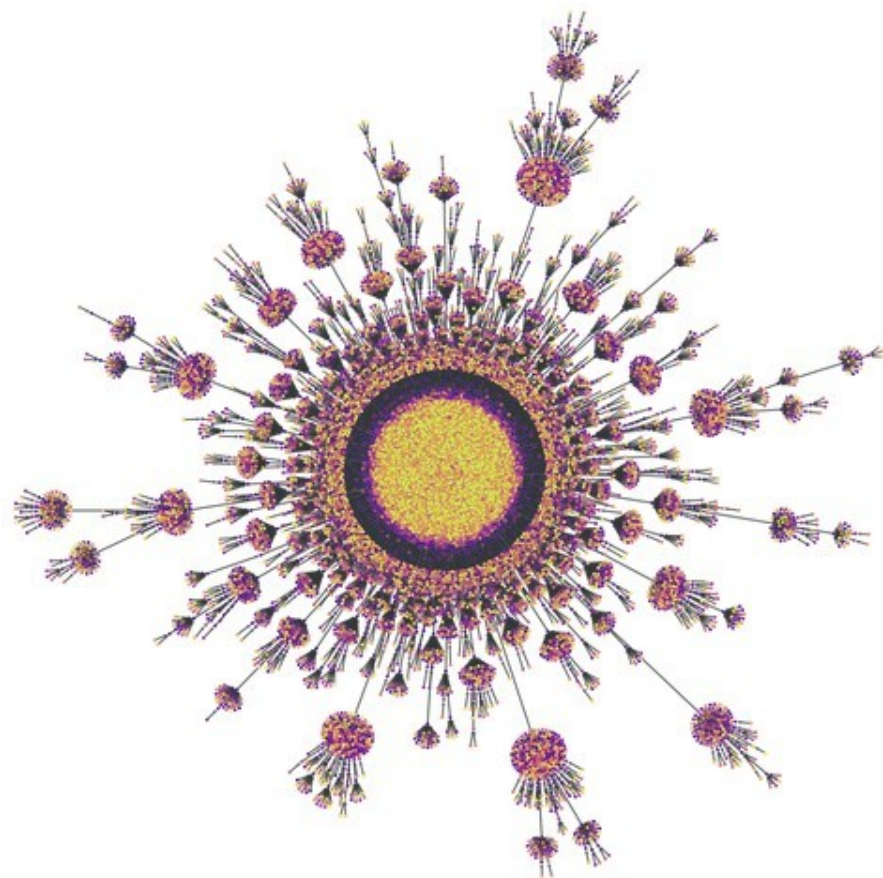
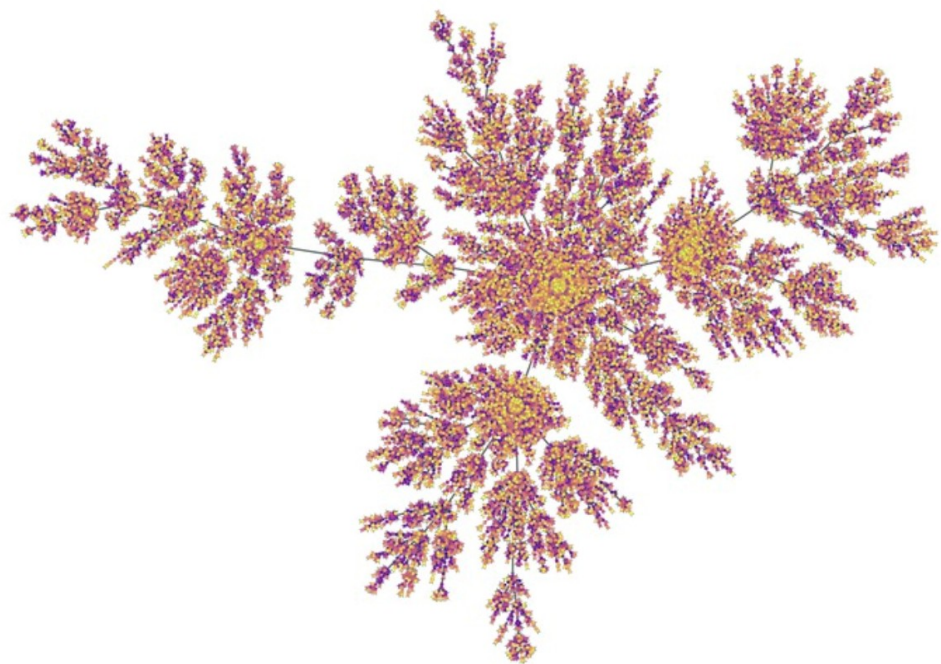








Nodes that join the network earlier have higher degree



Heavy-tailed degrees alone can fake clustering and communities

- A hub-dominated degree sequence can produce apparent clustering and apparent community structure with no real mechanism behind it
- That's exactly the boring explanation rung 2 (the configuration model) controls for

Configuration model

- More realistic than ER
- Edges are random conditional on a specified degree sequence
- Unlike the ER model, probability of the degree of a randomly selected node is not the same as the probability of the degree of its neighbours

The friendship paradox

Your friends are more popular than you are

The friendship paradox

Your friends are more popular than you are

Some people have **no friends**.

The friendship paradox

Your friends are more popular than you are

Some people have **no friends**. But because **they appear in nobody's friendship circles**, they're not making anyone else feel popular.

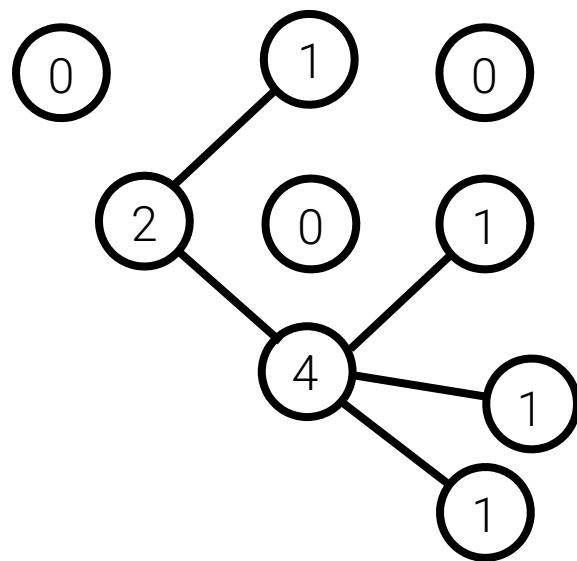


The friendship paradox

Your friends are more popular than you are

Some people have **no friends**. But because **they appear in nobody's friendship circles**, they're not making anyone else feel popular.

The same applies to other people: the more friends you have, the likelier you are to be represented in people's friendship circles.



The friendship paradox

Your friends are more popular than you are

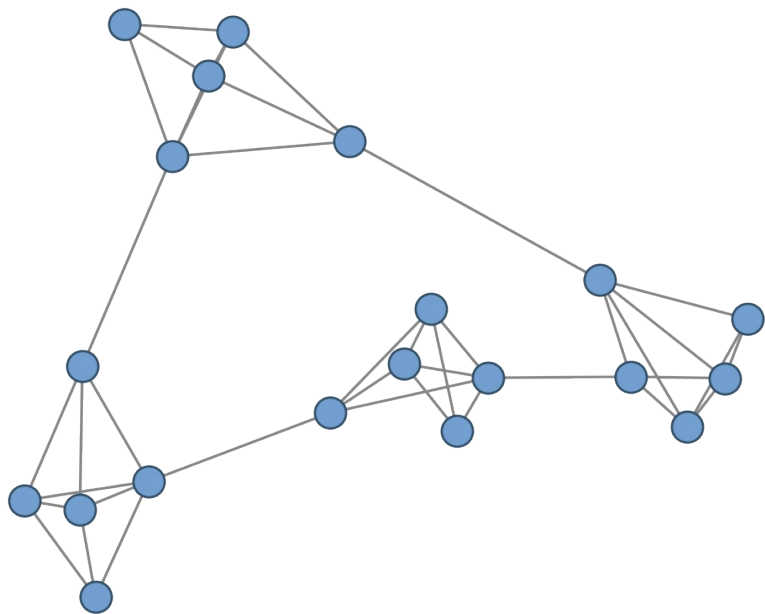
average friend
(count node proportional
to their degree)

average person
(count each node once)

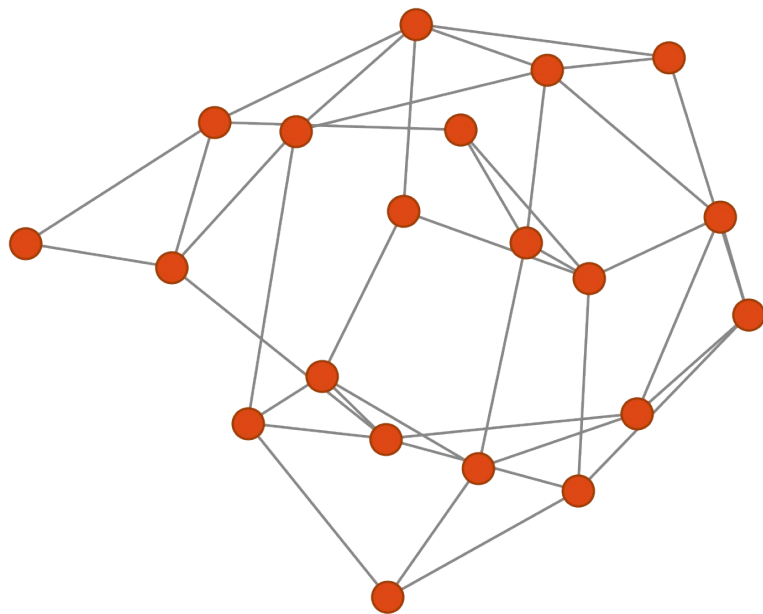
So popular people are oversampled as friends. Hence the paradox.

Same degrees, structure destroyed

Before: observed network
(clustered, communities visible)



After: configuration-model null
(same degree sequence, structure gone)



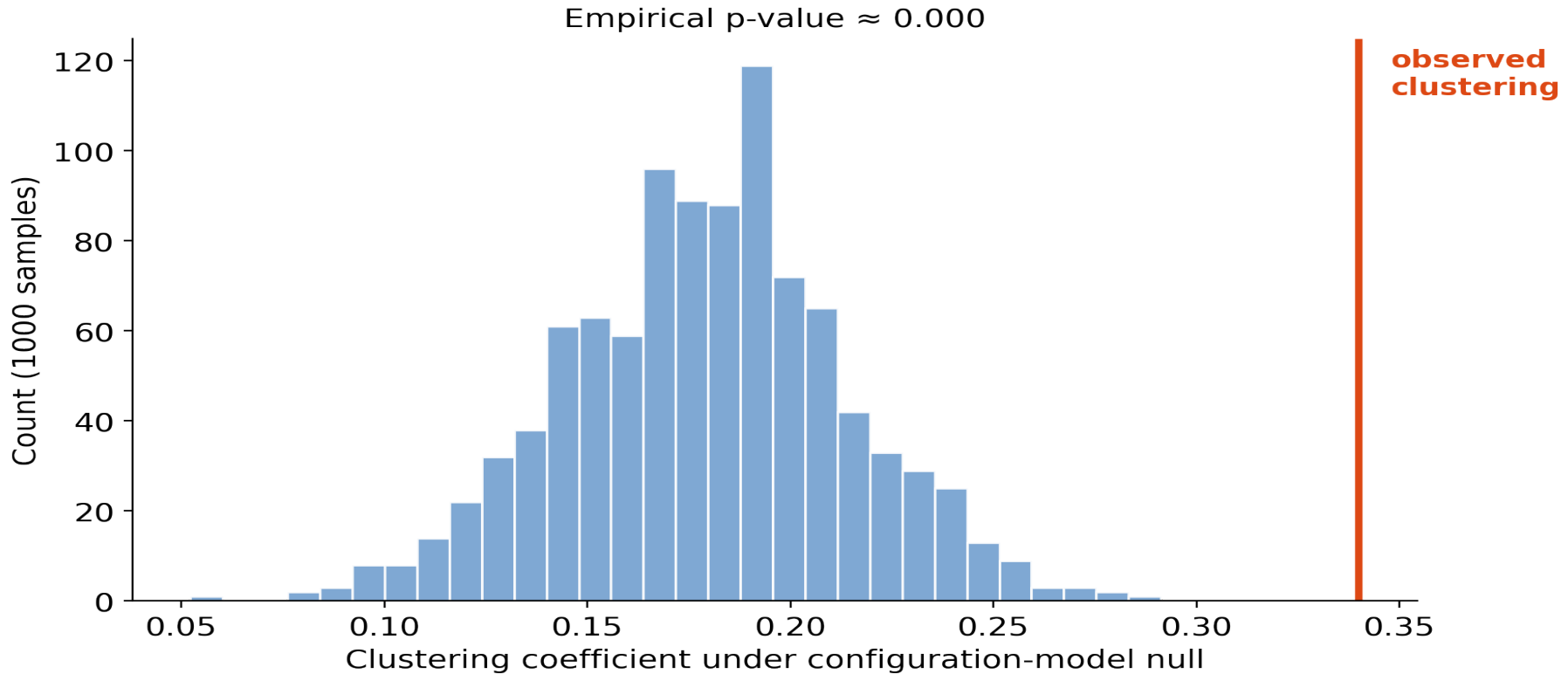
The configuration model preserves every node's degree but randomises everything else — whatever clustering or grouping survives that randomisation is worth explaining; whatever doesn't was just degree heterogeneity.

The four-step null-test recipe

1. Compute your statistic (e.g. clustering coefficient) on the observed network
2. Generate many configuration-model rewirings that preserve the degree sequence
3. Compute the same statistic on each rewiring → a null distribution
4. Compare the observed value to that distribution → empirical p-value

Same recipe, same variable names, reused Wednesday for the modularity null test.

Example: is the observed clustering surprising?

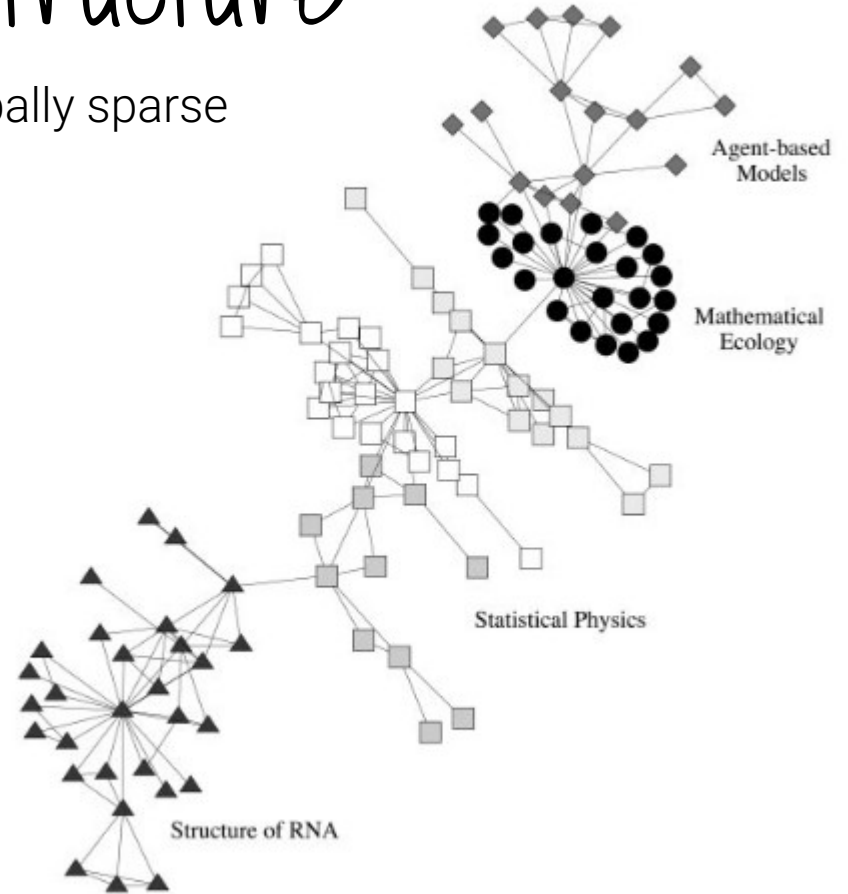
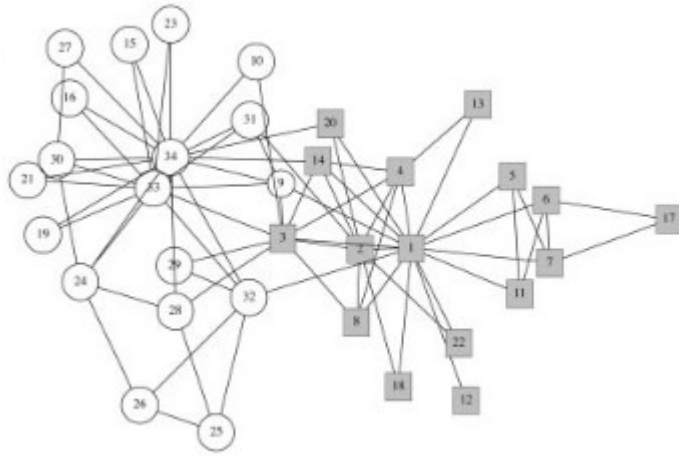


The null-model ladder – you are here

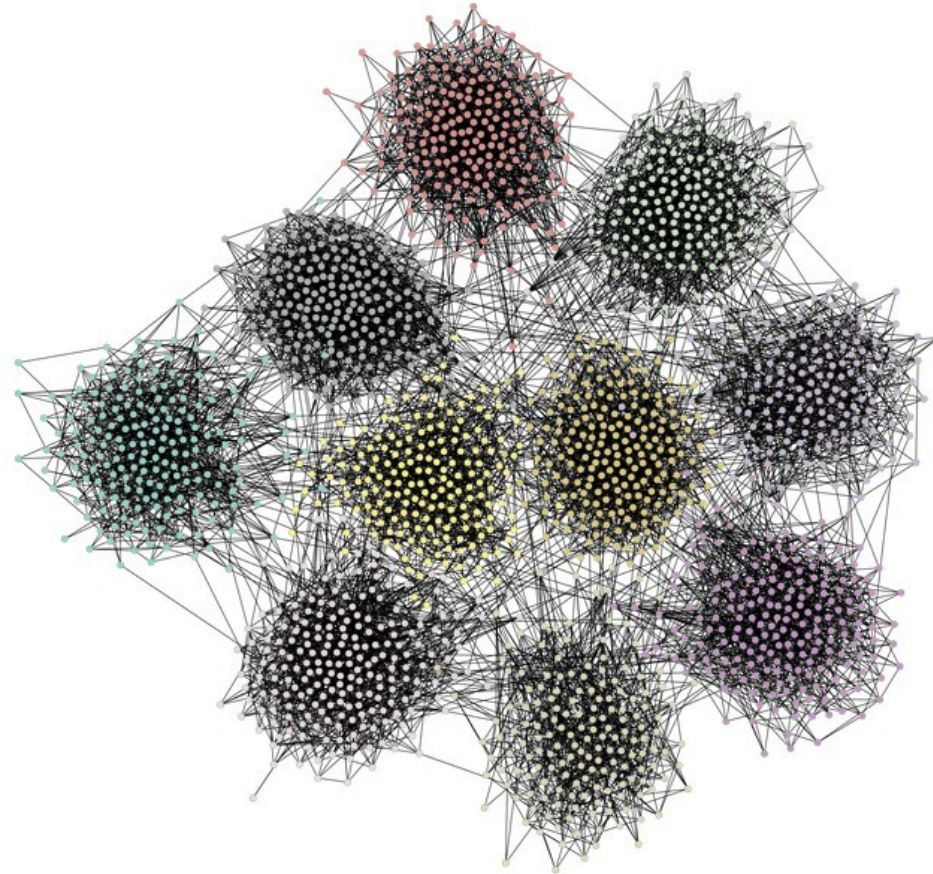
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Community structure

Locally dense, globally sparse



Stochastic Block Models

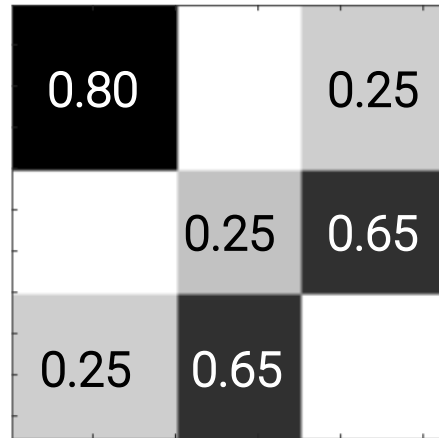


Generating a network using the SBM

Step 1 : Assign each node to a group

Generating a network using the SBM

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Mixing Matrix

- Step 2 : Select some connection probabilities (mixing matrix)

Generating a network using the SBM

Step 1 : Assign each node to a group

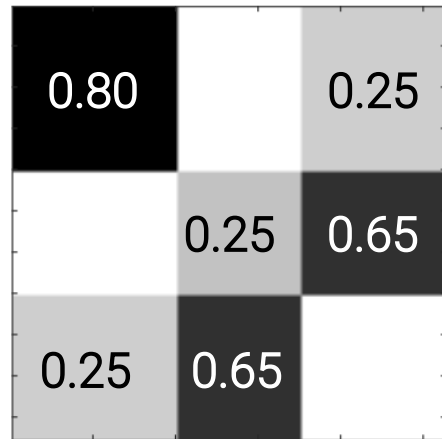


0.80		0.25
	0.25	0.65
0.25	0.65	

Mixing Matrix

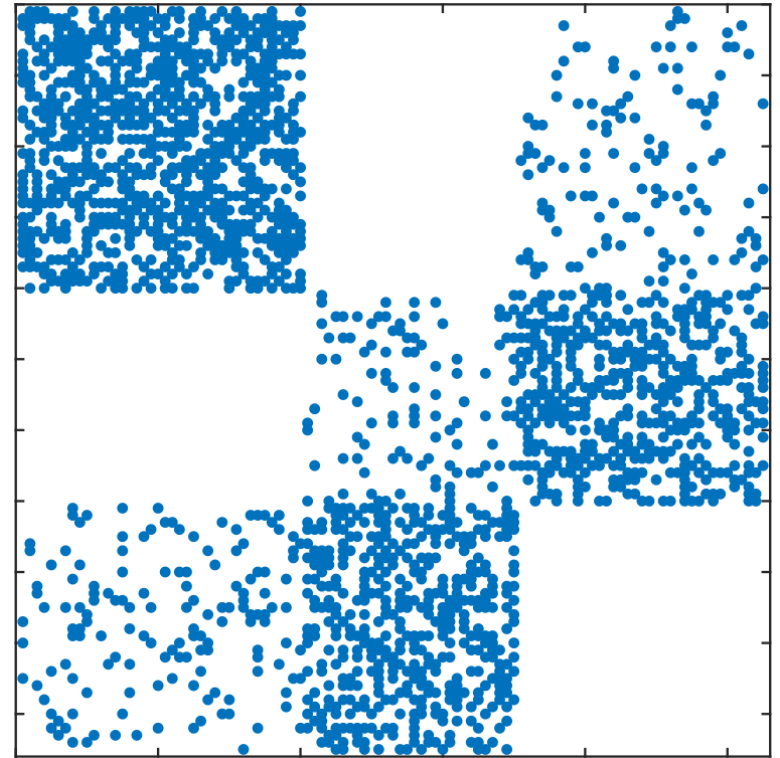
- *Step 2* : Select some connection probabilities (mixing matrix)
- *Step 3* : For each pair of nodes, add an edge with probability according to the group memberships

Generating a network using the SBM



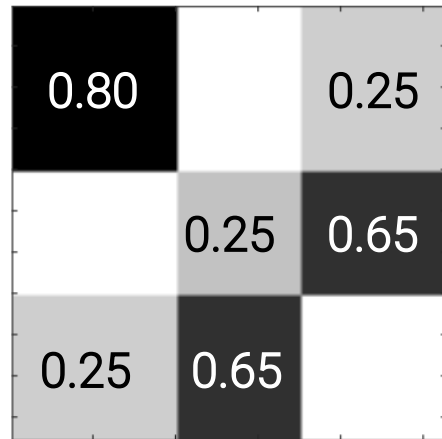
Mixing Matrix

generation
→



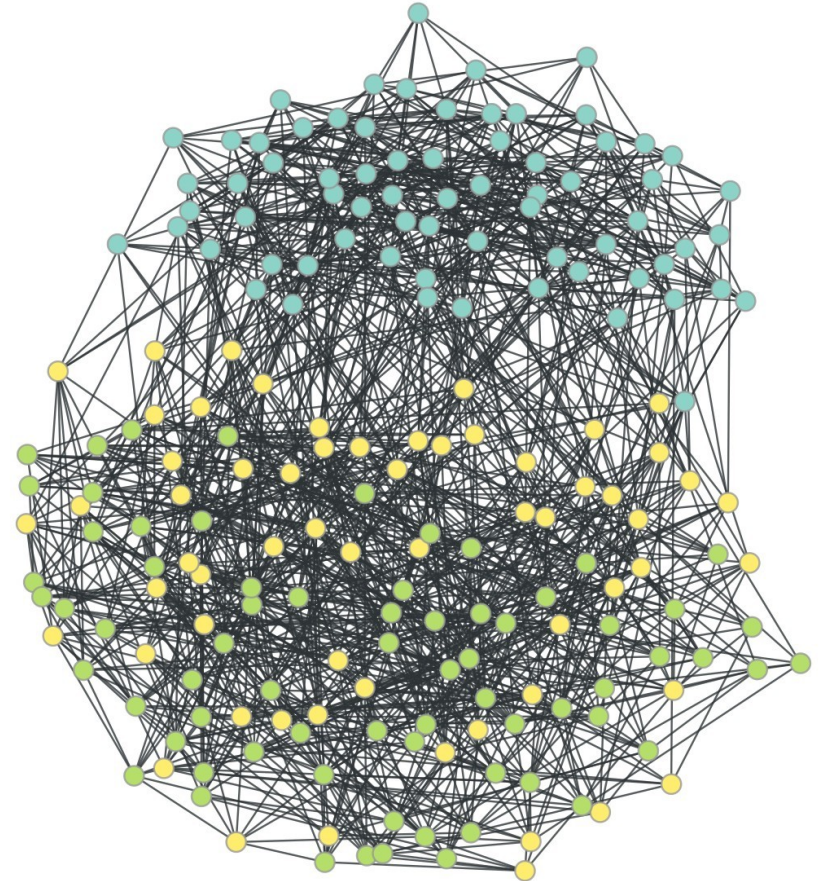
Adjacency Matrix

Generating a network using the SBM



Mixing Matrix

generation



Generating a network using the SBM

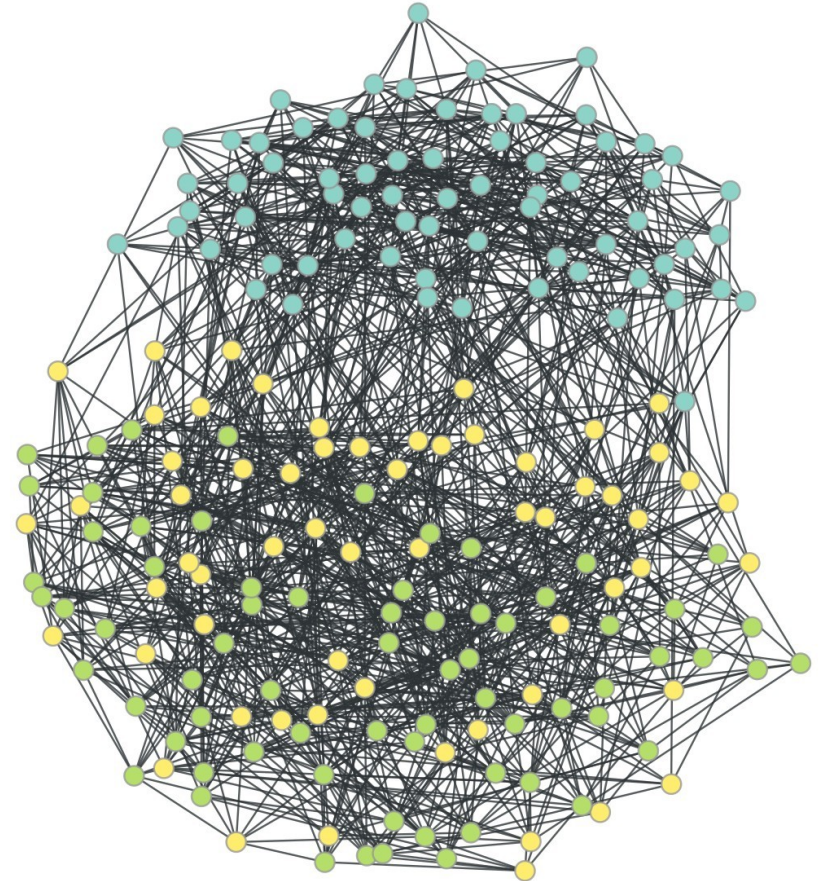
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Mixing Matrix

generation

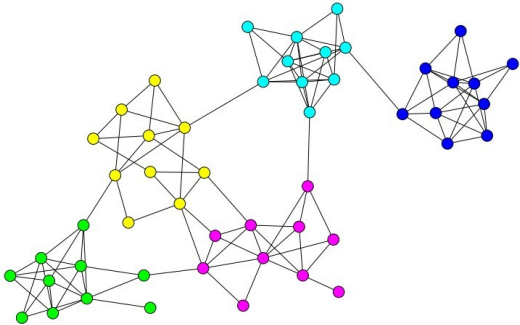
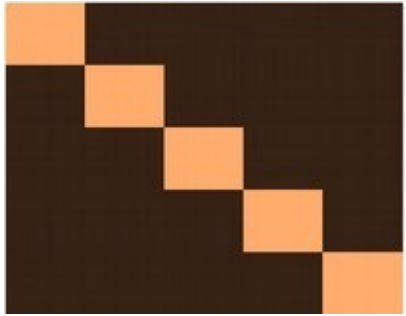


inference
(tomorrow)



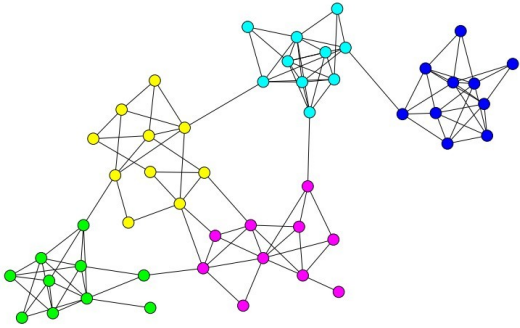
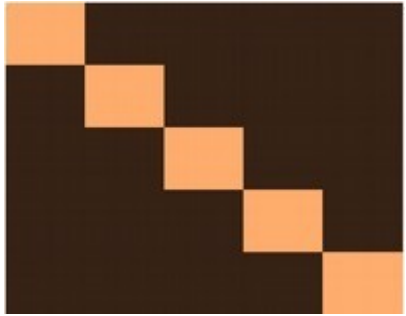
Different types of structure

assortative

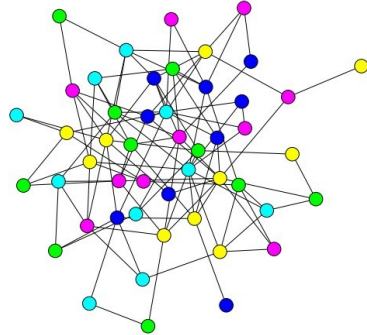
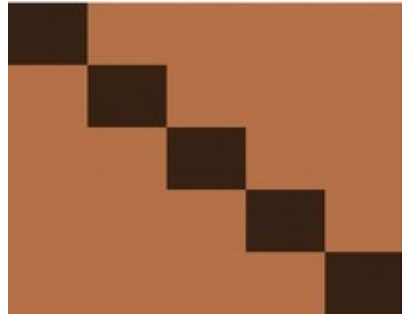


Different types of structure

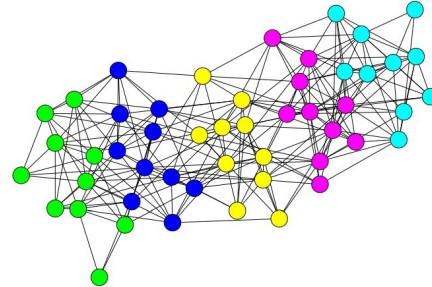
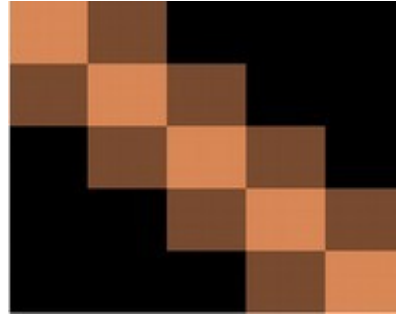
assortative



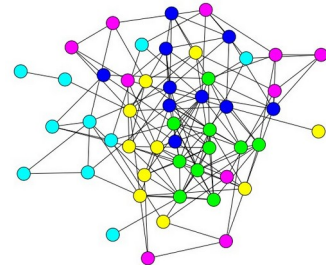
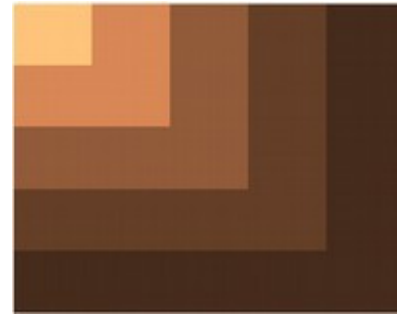
disassortative



ordered



core-periphery



DC-SBM: the one-liner to fit it

- Block structure exists as a model: nodes belong to groups, and edge probability depends on group membership (and degree, for the DC variant)
- Fitting it is one line of code — the `graph-tool` / equivalent `minimize` call you'll use again and again
- Today: just enough to know it's here and what it controls for. Wednesday owns the full treatment: generating, mixing matrices, structure types, model selection.

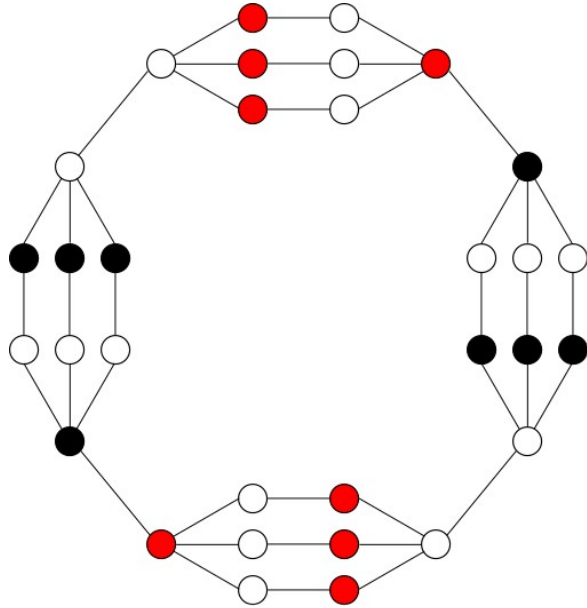
Entropy: how surprised the model is by the network

- Lower description length / entropy $\rightarrow$ the block structure explains the network well
- This is the intuition BESTest borrows next: how much does knowing an attribute reduce that surprise?

The null-model ladder – you are here

Rung	Null model	What it holds fixed	Failing it tells you...
1	Erdős–Rényi	number of nodes & edges	"not maximally random" (almost everything fails this)
2	Configuration model	every node's degree	there is structure beyond degree heterogeneity
3	DC-SBM	degrees and group structure	structure beyond blocks + degrees → Wednesday's session
—	BESTest	a node attribute you have	this attribute is (not) associated with the structure

Network nodes can have properties or attributes (metadata)



- Metadata (M) values
- Metadata (M) unknown

social networks *age, sex, ethnicity, race, etc.*

food webs *feeding mode, species body mass, etc.*

internet *data capacity, physical location, etc.*

protein interactions *molecular weight, association with cancer, etc.*

Blockmodel Entropy Significance Test

How well do the metadata explain the network?

Blockmodel Entropy Significance Test

How well do the metadata explain the network?

1. Divide the network G into groups according to metadata labels M .

Blockmodel Entropy Significance Test

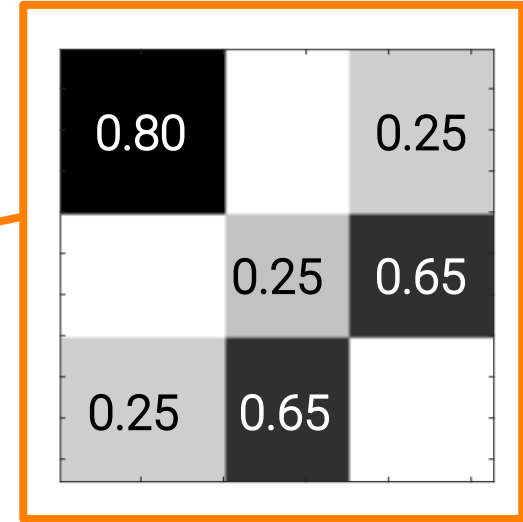
How well do the metadata explain the network?

1. Divide the network G into groups according to metadata labels M .
2. Fit the parameters of an SBM and compute the entropy $H(G, M)$

Blockmodel Entropy Significance Test

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Blockmodel Entropy Significance Test

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Test statistic



Blockmodel Entropy Significance Test

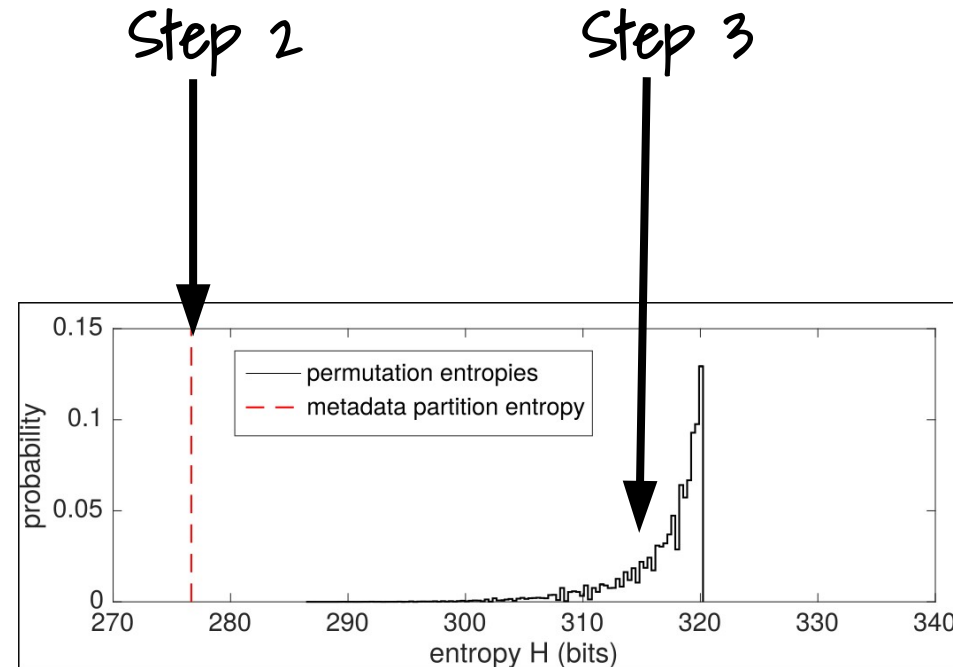
How well do the metadata explain the network?

1. Divide the network G into groups according to metadata labels M .
2. Fit the parameters of an SBM and compute the entropy $H(G, M)$
3. Compare this entropy to a distribution of entropies of networks partitioned using random permutations of the metadata labels.

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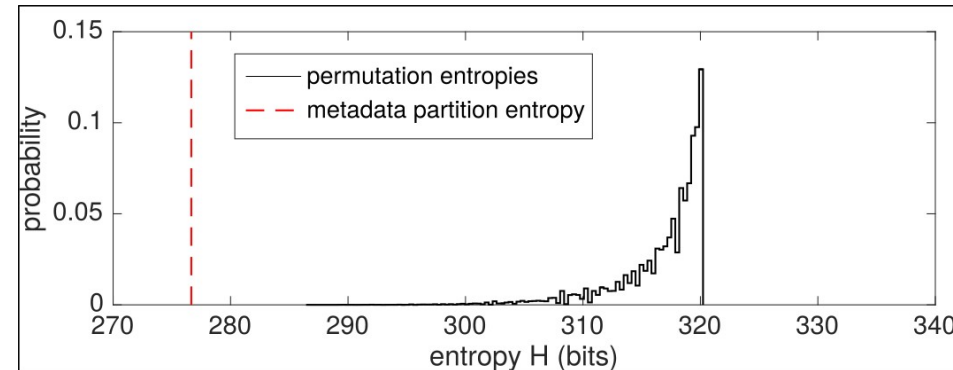
Blockmodel Entropy Significance Test

How well do the metadata explain the network?

1. Divide the network G into groups according to metadata labels M .
2. Fit the parameters of an SBM and compute the entropy $H(G, M)$
3. Compare this entropy to a distribution of entropies of networks partitioned using random permutations of the metadata labels.

metadata is randomly assigned
→ model gives no explanation, high H

metadata correlates with structure
→ model gives good explanation, low H

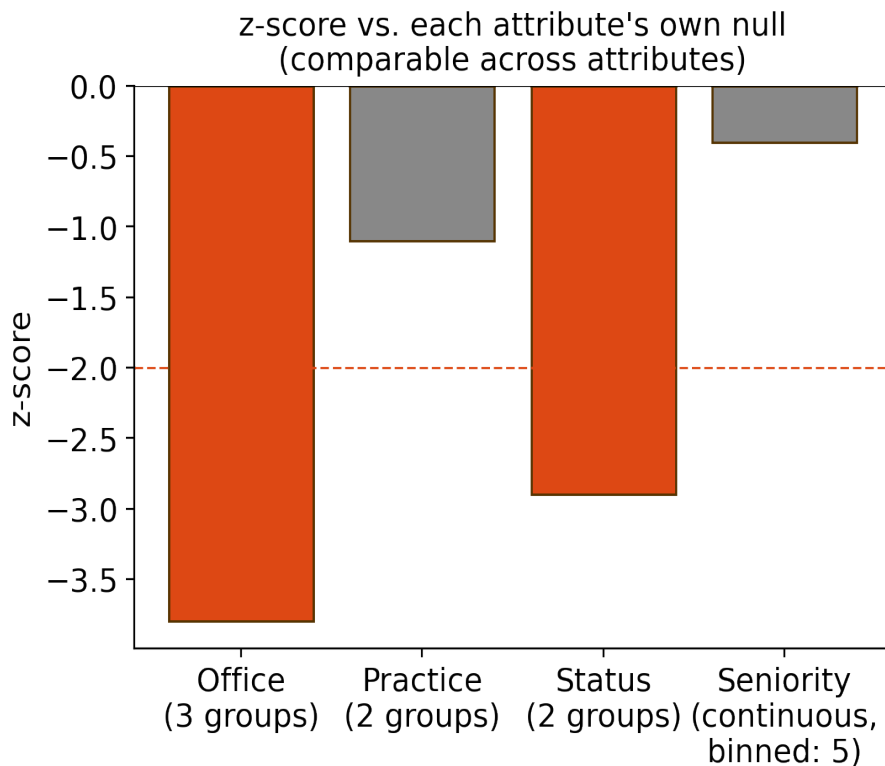
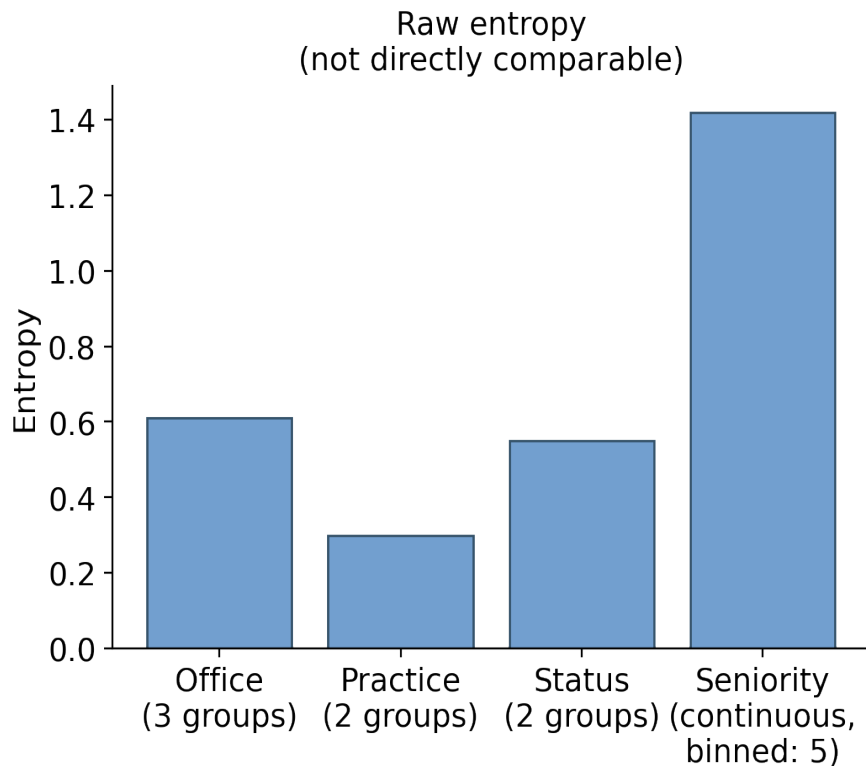


Multiple networks; multiple metadata attributes

Network	Status	Gender	Office	Practice	Law School
Friendship	$< 10^{-6}$	0.034	$< 10^{-6}$	0.033	0.134
Cowork	$< 10^{-3}$	0.094	$< 10^{-6}$	$< 10^{-6}$	0.922
Advice	$< 10^{-6}$	0.010	$< 10^{-6}$	$< 10^{-6}$	0.205

Multiple sets of metadata provide a significant explanation for multiple networks.

Comparing attributes fairly: z-scores, not raw entropy



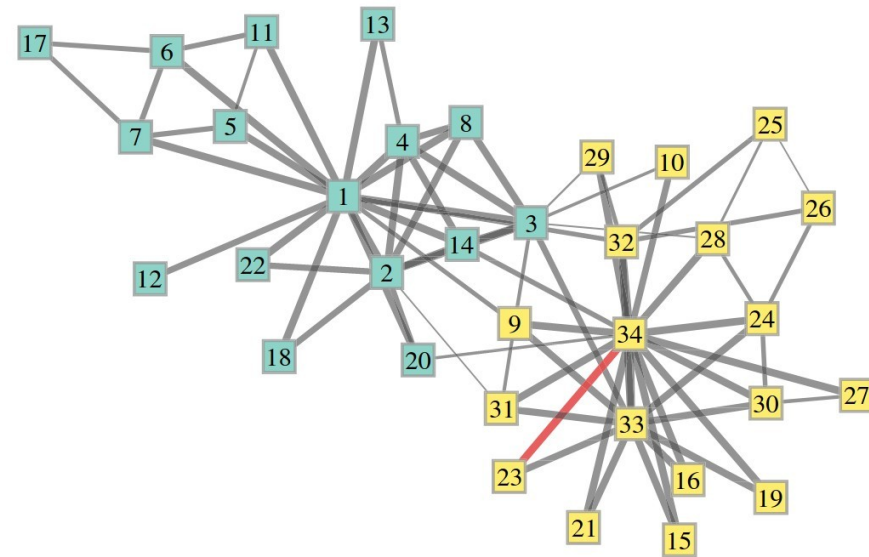
- Raw entropies aren't comparable across attributes with different group counts. Compare each attribute against its own null instead — the z-score is comparable.

Summary – the ladder, recapped

Rung	Null model	What it holds fixed	Failing it tells you...
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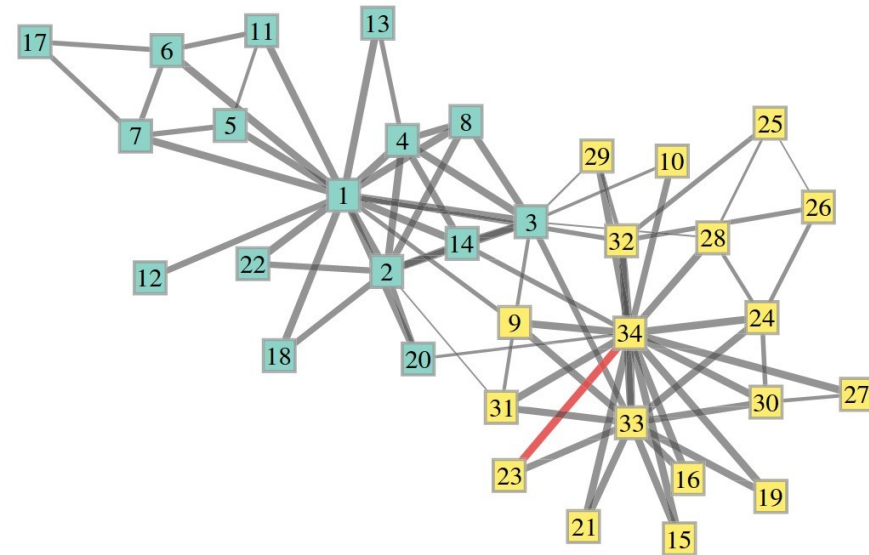
Errors in networks and reconstruction

Zachary's Karate Club



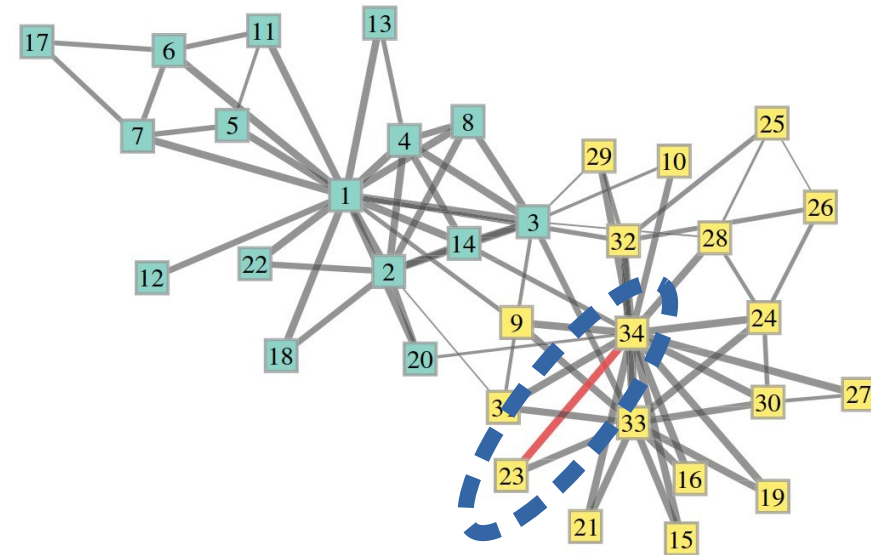
Zachary's Karate Club

	Individual Number																																		
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Zachary's Karate Club

	Individual Number																																			
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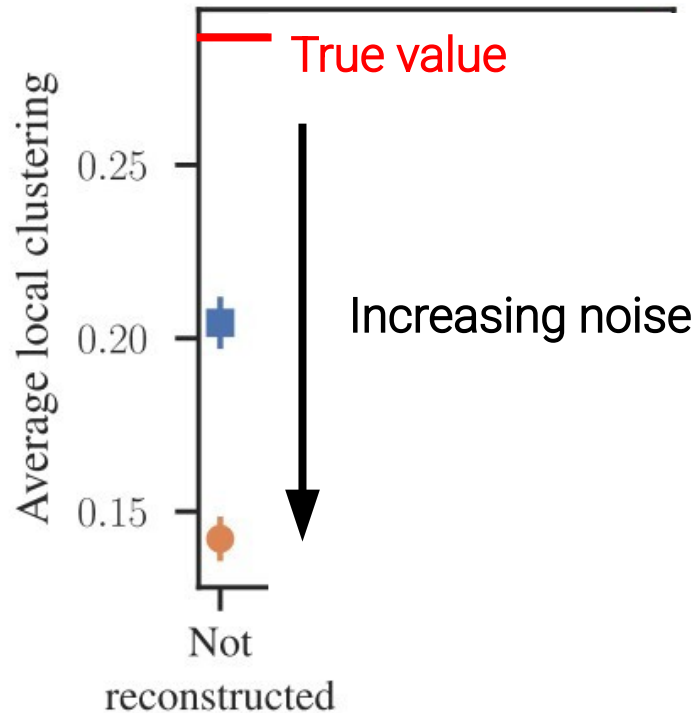


Does this edge exist?

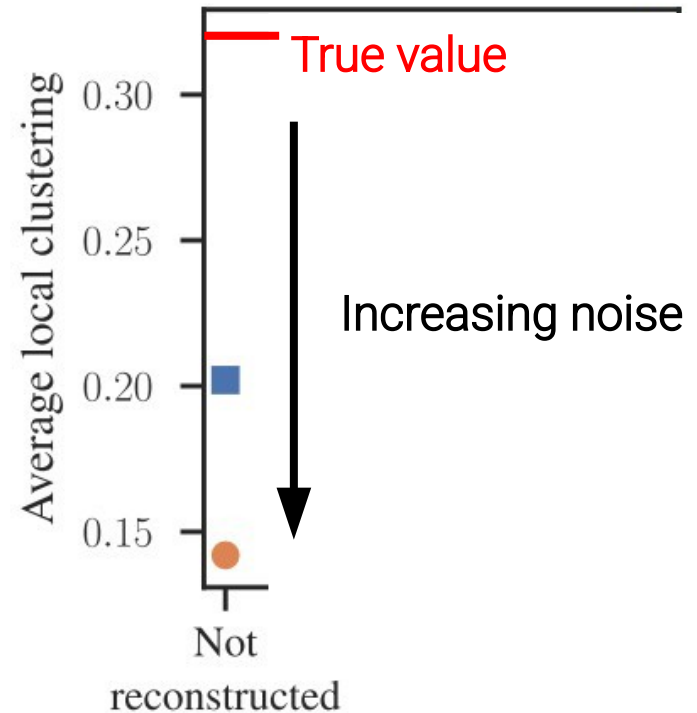
Errors in network data create systematic biases...

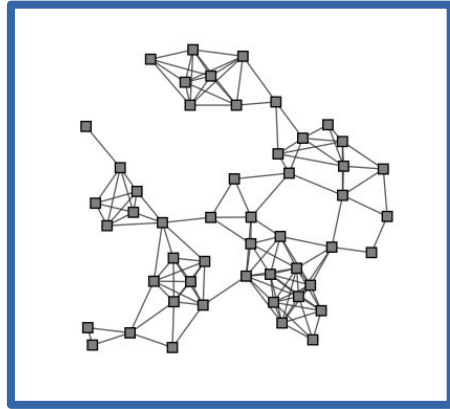
Errors in network data create systematic biases...

(a) High-school friendships

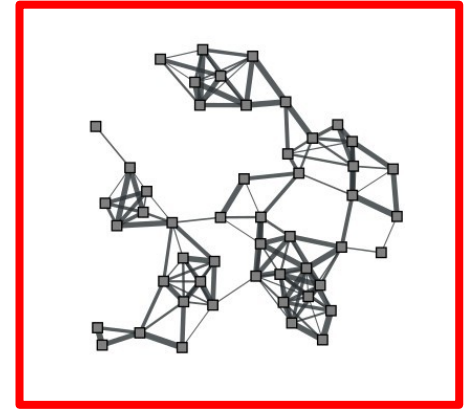


(b) Political blogs





True Network



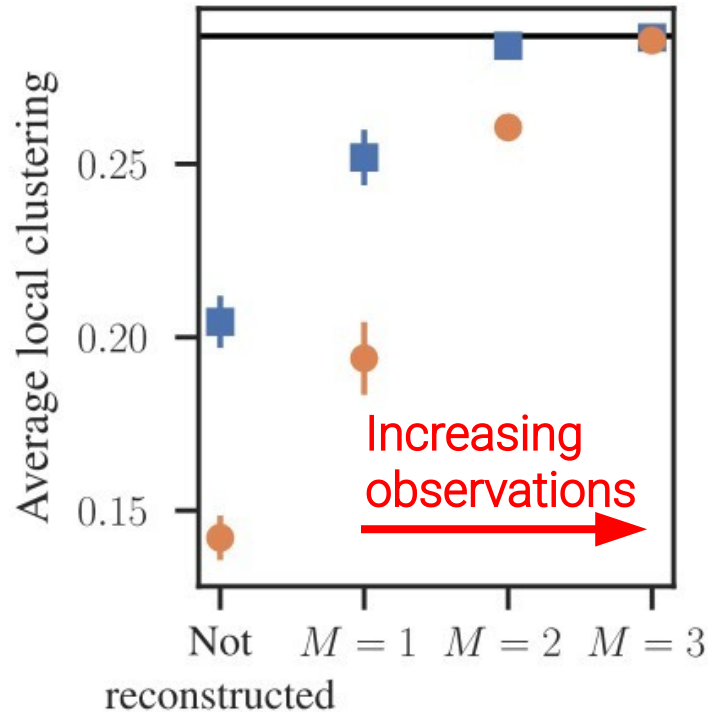
Reconstructed Network

We don't know if the network represents the system

$$P(\mathbf{A}|\mathbf{D}) = \frac{P(\mathbf{D}|\mathbf{A})P(\mathbf{A})}{P(\mathbf{D})} .$$

Bayesian inference

(a) High-school friendships



(b) Political blogs

